

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A150NRG10EM001-Final-ASM Review.pvdb

Date of Analysis : Sep 5, 2026 5:19pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2900.000 x 2900.000 mm ID	
Vessel Design Length, Tangent to Tangent	2853.00 cm
Distance of Bottom Tangent above Grade	700.00 cm
Distance of Base above Grade	700.00 cm
Specified Datum Line Distance	0.00 cm
Internal Design Temperature	290 °C
Internal Design Pressure	194.759 kgf/cm ²
External Design Temperature	0 °C
Maximum Allowable Working Pressure	196 kgf/cm ²
Shop Test Pressure	310.590 kgf/cm ²
Required Minimum Design Metal Temperature	-10.0 °C
Warmest Computed Minimum Design Metal Temperature	-4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	+IT
Head	SA-387 11	2	...	K11789	+IT
Flange	SA-336 F11	3	...	K11572	+IT
Skirt	SA-387 11	2	...	K11789	+IT
Nozzle	SA-336 F11	3	...	K11572	+IT
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-336 F11	3	...	K11572	
Basering	SA-387 11	2	...	K11789	
Flg Bolting	SA-540 B23	3	<= 240	H43400	
Base Bolting	SA-193 B7M	...	<= 100	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom head	194.759	0.00	199.000	1.5000	N/A	No
Shell-1	194.759	0.00	196.000	3.0000	N/A	No
Skirt	...	...	...	3.0000	N/A	No
Shell-2	194.759	0.00	198.000	1.5000	N/A	No
Shell-3	194.759	0.00	198.000	1.5000	N/A	No
Shell-4	194.759	0.00	198.000	1.5000	N/A	No
Shell-5	194.759	0.00	198.000	1.5000	N/A	No
Shell-6	194.759	0.00	198.000	1.5000	N/A	No
Shell-7	194.759	0.00	198.000	1.5000	N/A	No
Shell-8	194.759	0.00	198.000	1.5000	N/A	No
Shell-9	194.759	0.00	198.000	1.5000	N/A	No
Shell flange	194.759	0.00	198.000	1.5000	N/A	No
Top cover flang	194.759	0.00	197.000	1.5000	N/A	No

Element Types and Properties:



Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Sphere	0.0	0.0	107.0	93.0	90.8	...	1.00	1.00
Cylinder	435.0	435.0	183.0	183.0	181.5	...	1.00	1.00
Skirt	435.0	300.0	60.0	60.0	...	...	1.00	1.00
Cylinder	600.0	165.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	900.0	300.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	1200.0	300.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	1500.0	300.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	1800.0	300.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	2100.0	300.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	2400.0	300.0	183.0	183.0	179.9	...	1.00	1.00
Cylinder	2680.2	280.2	183.0	183.0	179.9	...	1.00	1.00
Body Flg	2808.0	127.5	1000.0	1000.0	410.7	408.6	1.00	1.00
Body Flg	2853.0	45.0	700.0	700.0	695.5	368.6	1.00	1.00

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	226.462	...	...	...	...
20	30	82.5	642.679	0.0070007	587.81	0.006403
30	40	150	21344.9	186158	255183	3410372
40	50	382.5	16378	180084	202685	2864115
50	60	615	15359	174352	176502	2571705
60	70	915	13454.8	164131	133281	2063980
70	80	1215	11492.8	151829	95859.8	1590038
80	90	1515	9480.15	136682	64400.3	1157271
90	100	1815	7422.25	115301	39046.7	779296
100	110	2115	5323.38	94161.6	19928.2	465102
110	120	2405.09	3187.02	71137.3	7162.63	217153
120	130	2608.93	1160.52	46196.6	1068.47	52633.4
130	140	2695.18	381.033	26879	85.7326	6047.78

Abs Max of the all of the Stress Ratio's : 0.9203

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	44.0000	mm
Inside Diameter of Basing	3346.0000	mm
Outside Diameter of Basing	4054.0002	mm
Nominal Diameter of Bolts	48.0000	mm
Diameter of Bolt Circle	3925.9998	mm
Number of Bolts	40	

Thickness of Gusset Plates	42.0000	mm
Average Width of Gusset Plates	134.0000	mm
Height of Gussets	197.0000	mm
Distance between Gussets	102.0000	mm
Thickness of Top Plate or Ring	39.0000	mm
Radial Width of the Top Plate	134.0000	mm
Number of Gussets	80	

Loads for Foundation/Support Design:

Total Wind Shear on Support	21345.	kgf
Total Earthquake Shear on Support	186158.	kgf
Wind Moment on Support	257058.	kgf-m.
Earthquake Moment on Support	3412247.	kgf-m.
Maximum Initial Bolt Load	63481.	kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included



are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated	- Bare w/o removable internals	591118.3	kg
Shop Test	- Fabricated + water (full)	783510.7	kg
Shipping	- Fab. + rem. intls.+ shipping app.	979898.2	kg
Erected	- Fab. + rem. intls.+ insul. (etc)	979898.2	kg
Empty	- Fab. + intls. + details + wghts.	979898.2	kg
Operating	- Empty + operating liquid (no ca)	979898.2	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
A2 12"	199	...	...	OK	OK	Passed
A1 12"	199	...	...	OK	OK	Passed
T11 2"	198	...	...	OK	OK	Passed
T12 2"	198	...	...	OK	OK	Passed
T13 2"	198	...	...	OK	OK	Passed
T8 2"	198	...	...	OK	OK	Passed
T9 2"	198	...	...	OK	OK	Passed
T10 2"	198	...	...	OK	OK	Passed
T5 2"	198	...	...	OK	OK	Passed
T6 2"	198	...	...	OK	OK	Passed
T7 2"	198	...	...	OK	OK	Passed
T1 2"	553	...	...	OK	OK	Passed
T2 2"	553	...	...	OK	OK	Passed
T3 2"	553	...	...	OK	OK	Passed
T4 2"	553	...	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 198 Nozzle : T7 2"
 Minimum MAWP Shells/Flanges : 196 Element : Shell-1
 Minimum MAPnc Shells/Flanges : 198 Element : Shell flange

Computed Vessel M.A.W.P. : 196 kgf/cm²

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
10	A2 12"	0.000	0.000	151.500
10	A1 12"	0.000	180.000	151.500
40	T11 2"	4550.000	0.000	26.500
40	T12 2"	4550.000	45.000	26.500
40	T13 2"	4550.000	90.000	26.500
70	T8 2"	14500.000	0.000	26.500
70	T9 2"	14500.000	45.000	26.500
70	T10 2"	14500.000	90.000	26.500
110	T5 2"	26000.004	0.000	26.500
110	T6 2"	26000.004	45.000	26.500
110	T7 2"	26000.004	90.000	26.500
130	T1 2"	0.000	0.000	26.500
130	T2 2"	0.000	90.000	26.500
130	T3 2"	0.000	180.000	26.500
130	T4 2"	0.000	270.000	26.500

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite 28 U2 Licensee: SL User

FileName : A150NRG10EM001-Final-ASM Review

Nozzle Summary:

Step:

49

5:19pm

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HEXAGON

No Multiple Nozzle spacing violations detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
T11 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T12 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T13 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T8 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T9 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T10 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T5 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T6 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T7 2"	50 mm	Actual	WNF	66.0	8.000	...	...	484	...
T1 2"	50 mm	Actual	WNF	66.0	8.000	...	...	1000	...
T2 2"	50 mm	Actual	WNF	66.0	8.000	...	...	1000	...
T3 2"	50 mm	Actual	WNF	66.0	8.000	...	...	1000	...
T4 2"	50 mm	Actual	WNF	66.0	8.000	...	...	1000	...
A2 12"	300 mm	Actual	None	344.0	22.000	...	...	610	...
A1 12"	300 mm	Actual	None	344.0	22.000	...	...	610	...

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T11 2"	SA-336 F11	183.000	10.000	...	...	...
T12 2"	SA-336 F11	183.000	10.000	...	...	...
T13 2"	SA-336 F11	183.000	10.000	...	...	...
T8 2"	SA-336 F11	183.000	10.000	...	...	...
T9 2"	SA-336 F11	183.000	10.000	...	...	...
T10 2"	SA-336 F11	183.000	10.000	...	...	...
T5 2"	SA-336 F11	183.000	10.000	...	...	...
T6 2"	SA-336 F11	183.000	10.000	...	...	...
T7 2"	SA-336 F11	183.000	10.000	...	...	...
T1 2"	SA-336 F11	700.000	10.000	...	...	...
T2 2"	SA-336 F11	700.000	10.000	...	...	...
T3 2"	SA-336 F11	700.000	10.000	...	...	...
T4 2"	SA-336 F11	700.000	10.000	...	...	...
A2 12"	SA-336 F11	93.000	10.000	...	...	...
A1 12"	SA-336 F11	93.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
T11 2"	455.000	0.0	300.00	0.00	Shell-2
T12 2"	455.000	45.0	300.00	0.00	Shell-2
T13 2"	455.000	90.0	300.00	0.00	Shell-2
T8 2"	1450.000	0.0	300.00	0.00	Shell-5
T9 2"	1450.000	45.0	300.00	0.00	Shell-5
T10 2"	1450.000	90.0	300.00	0.00	Shell-5
T5 2"	2600.000	0.0	300.00	0.00	Shell-9



T6 2"	2600.000	45.0	300.00	0.00	Shell-9
T7 2"	2600.000	90.0	300.00	0.00	Shell-9
T1 2"	...	0.0	300.00	0.00	Top cover flang
T2 2"	...	90.0	300.00	0.00	Top cover flang
T3 2"	...	180.0	300.00	0.00	Top cover flang
T4 2"	...	270.0	300.00	0.00	Top cover flang
A2 12"	...	0.0	500.00	0.00	Bottom head
A1 12"	...	180.0	500.00	0.00	Bottom head



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 107.0mm THK X 2992.0mm ID X 0.0cm	SA-387 11
1	WELD NECK FLANGE: 1000.0mm THK X 4000.0mm OD	SA-336 F11
1	BLIND FLANGE: 700.0mm THK X 4000.0mm OD	SA-336 F11
2	PACKING: 4350.000mm	...
1	INSULATION: 1300mm X 100mm THK	
2	PACKING: 1650.000mm	...
1	INSULATION: 1650mm X 100mm THK	
10	PACKING: 3000.000mm	...
6	INSULATION: 3000mm X 100mm THK	
1	PACKING: 1375.000mm	...
1	PACKING: 1000.000mm	...
1	PACKING: 2335.000mm	...
1	PACKING: 500.000mm	...
1	PACKING: 2520.000mm	...
1	PACKING: 2801.800mm	...
1	INSULATION: 2801mm X 100mm THK	
1	BASERING: 4054.0mm OD X 3346.0 ID X 44.0mm	SA-387 11
40	BASERING BOLTS: 48.0mm DIA	SA-193 B7M
40	BASERING BOLT NUTS: 48.0mm DIA	...
1	BASERING TOP RING: 3653.7mm OD X 3385.7 ID X 39.0mm	...
80	BASERING GUSSET PLATES: 134.0mm X 134.0mm X 197.0mm X 42.0mm	...
1	GASKET: 3010.0mm OD X 2930.0mm ID	...
40	BODY FLANGE BOLTS: 150.0mm DIA	SA-540 B23
80	NUTS FOR BODY FLANGE BOLTS: 150.0mm DIA	...
2	CYLINDRICAL SEGMENT 183.0mm THK X 3624.9mm X 435.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 183.0mm THK X 1860.6mm X 435.0cm	SA-387 11
2	CON SKIRT SEGMENT 60.0mm THK X 3432.3mm X 300.0cm X 3852.8mm	SA-387 11
1	CON SKIRT SEGMENT 60.0mm THK X 3394.7mm X 300.0cm X 3810.5mm	SA-387 11
2	CYLINDRICAL SEGMENT 183.0mm THK X 3048.0mm X 165.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 183.0mm THK X 3014.6mm X 165.0cm	SA-387 11
7	CYLINDRICAL SEGMENT 183.0mm THK X 3048.0mm X 300.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 183.0mm THK X 3014.6mm X 300.0cm	SA-387 11
5	CYLINDRICAL SEGMENT 183.0mm THK X 6062.5mm X 300.0cm	SA-387 11



5	CYLINDRICAL SEGMENT 183.0mm THK X 0.0mm X 300.0cm	SA-387 11
1	CYLINDRICAL SEGMENT 183.0mm THK X 6062.5mm X 280.2cm	SA-387 11
1	CYLINDRICAL SEGMENT 183.0mm THK X 3048.0mm X 280.2cm	SA-387 11
1	CYLINDRICAL SEGMENT 183.0mm THK X 0.0mm X 280.2cm	SA-387 11
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
Shell flange [11]	!	-10		No			
Top cover fla[11]	!	-10		No			
Bottom head [9]	!	-10		No			
Shell-1 [8]	!	-10		No			
Shell-2 [8]	!	-10		No			
Shell-3 [8]	!	-10		No			
Shell-4 [8]	!	-10		No			
Shell-5 [8]	!	-10		No			
Shell-6 [8]	!	-10		No			
Shell-7 [8]	!	-10		No			
Shell-8 [8]	!	-10		No			
Shell-9 [8]	!	-10		No			
A2 12" [1]	!	-10		No			
A1 12" [1]	!	-10		No			
T11 2" [1]	!	21	-4	No	0.394	8.000	1.00
T12 2" [1]	!	21	-4	No	0.394	8.000	1.00
T13 2" [1]	!	21	-4	No	0.394	8.000	1.00
T8 2" [1]	!	21	-4	No	0.394	8.000	1.00
T9 2" [1]	!	21	-4	No	0.394	8.000	1.00
T10 2" [1]	!	21	-4	No	0.394	8.000	1.00
T5 2" [1]	!	-10		No			
T6 2" [1]	!	-10		No			
T7 2" [1]	!	-10		No			
T1 2" [1]	!	-10		No			
T2 2" [1]	!	-10		No			
T3 2" [1]	!	-10		No			
T4 2" [1]	!	-10		No			
Nozl. Bolting[22]		-48		No	1.000	64.000	1.00

Warmest MDMT: 21 -4

Required Minimum Design Metal Temperature -10.0 °C

Warmest Computed Minimum Design Metal Temperature -4.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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**Units used in this Analysis (Kgfc^{m2}):**

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in. ⁴	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in. ³	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	194.76	kgf/cm ²
Design Internal Temperature	290.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-10.0	°C
Type of Construction	Welded	
Special Service	None	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	User Defined Hydro
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW



Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	0	
Base Elevation	700	cm
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh 0	cm
Distance Upwind of Crest	Lh 0	cm
Distance from Crest to the Vessel	x 0	cm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	cm

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	1.500	
Site Class	D	
Component Elevation Ratio	z/h 0.000	
Amplification Factor	Ap 2.500	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Component resonance duct. fact.	[CAR] 0.000	
Component strength factor	[Rpo] 0.000	
Overstrength factor	[Omega0] 0.000	
Response Acceleration Factor	[SMS] 0.326	
Response Acceleration Factor	[SM1] 0.139	
Building Importance Factor	[Ie] 1.000	
Site Latitude	0.000	deg
Site Longitude	0.000	deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/13
Element From Node	10
Element To Node	20
Element Type	Spherical
Description	Bottom head
Distance "FROM" to "TO"	0 cm
Inside Diameter	2992 mm
Element Thickness	93 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	107 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.76 kgf/cm²
Design Temperature Internal Pressure	290 °C



Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Allowable Stress, Ambient	2110.7	kgf/cm ²
Allowable Stress, Operating	1680.4	kgf/cm ²
Allowable Stress, Hydrotest	3002.9	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2520.6	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	10	
Detail Type	Nozzle	
Detail ID	A2 12"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	300	mm
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	255.12	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

Element From Node	10	
Detail Type	Nozzle	
Detail ID	A1 12"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	300	mm
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	255.12	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

Element From Node	10	
Detail Type	Weight	
Detail ID	Bottom assembly	
Dist. from "FROM" Node / Offset dist	-140	cm
Miscellaneous Weight	20000	kgf
Offset from Element Centerline	0	mm

Element#	2/13	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	435	cm
Inside Diameter	2900	mm
Element Thickness	183	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	183	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	194.76	kgf/cm ²
Design Temperature Internal Pressure	290	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	



Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Packing
Detail ID	Bed-3 1
Dist. from "FROM" Node / Offset dist	0 cm
Height of Packed Section	435 cm
Density	0.00218 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0
Element From Node	20
Detail Type	Packing
Detail ID	Basket
Dist. from "FROM" Node / Offset dist	0 cm
Height of Packed Section	435 cm
Density	0.000649 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0
Element From Node	20
Detail Type	Insulation
Detail ID	IN
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	130 cm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm ³
Element From Node	20
Detail Type	Weight
Detail ID	SF weight
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	3749.9 kgf
Offset from Element Centerline	1500 mm
Element From Node	20
Detail Type	Weight
Detail ID	Sup ring wt
Dist. from "FROM" Node / Offset dist	0 cm
Miscellaneous Weight	7000 kgf
Offset from Element Centerline	1500 mm

Element#	3/13
Element From Node	30
Element To Node	40
Element Type	Skirt Support
Description	Skirt
Distance "FROM" to "TO"	300 cm
Skirt Outside Diameter	3386 mm
Diameter of Skirt at Base	3786 mm
Skirt Thickness	60 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	60 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	290 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Head-to-Skirt or Circ. Seam	1.0
Weld is pre-Heated	No



Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	165 cm
Inside Diameter	2900 mm
Element Thickness	183 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	183 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.76 kgf/cm ²
Design Temperature Internal Pressure	290 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Packing
Detail ID	Bed-3 2
Dist. from "FROM" Node / Offset dist	0 cm
Height of Packed Section	165 cm
Density	0.00218 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	40
Detail Type	Packing
Detail ID	Basket
Dist. from "FROM" Node / Offset dist	0 cm
Height of Packed Section	165 cm
Density	0.000649 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	40
Detail Type	Insulation
Detail ID	INS
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	165 cm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm ³

Element From Node	40
Detail Type	Nozzle
Detail ID	T11 2"
Dist. from "FROM" Node / Offset dist	20 cm
Nozzle Diameter	50 mm
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	50.906 kgf
Grade of Attached Flange	None
Nozzle Matl	SA-336 F11 +IT

Element From Node	40
Detail Type	Nozzle
Detail ID	T12 2"
Dist. from "FROM" Node / Offset dist	20 cm
Nozzle Diameter	50 mm
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	45.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	50.906 kgf
Grade of Attached Flange	None



Nozzle Matl SA-336 F11 +IT

Element From Node 40
 Detail Type Nozzle
 Detail ID T13 2"
 Dist. from "FROM" Node / Offset dist 20 cm
 Nozzle Diameter 50 mm
 Nozzle Schedule None
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 50.906 kgf
 Grade of Attached Flange None
 Nozzle Matl SA-336 F11 +IT

Element From Node 40
 Detail Type Weight
 Detail ID Skirt wt
 Dist. from "FROM" Node / Offset dist 100 cm
 Miscellaneous Weight 25600 kgf
 Offset from Element Centerline 0 mm

Element# 5/13
 Element From Node 50
 Element To Node 60
 Element Type Cylinder
 Description Shell-3
 Distance "FROM" to "TO" 300 cm
 Inside Diameter 2900 mm
 Element Thickness 183 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 183 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 194.76 kgf/cm²
 Design Temperature Internal Pressure 290 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-387 11 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Packing
 Detail ID Bed-3 3
 Dist. from "FROM" Node / Offset dist 0 cm
 Height of Packed Section 300 cm
 Density 0.00218 kg/cm³
 Percent Volume Holdup 0.0
 Specific Gravity of Packing Liquid 0.0

Element From Node 50
 Detail Type Packing
 Detail ID Basket
 Dist. from "FROM" Node / Offset dist 0 cm
 Height of Packed Section 300 cm
 Density 0.000649 kg/cm³
 Percent Volume Holdup 0.0
 Specific Gravity of Packing Liquid 0.0

Element From Node 50
 Detail Type Insulation
 Detail ID Ins: 50
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 300 cm
 Thickness of Insulation 100 mm
 Density 0.0002 kg/cm³



Element# 6/13
 Element From Node 60
 Element To Node 70
 Element Type Cylinder
 Description Shell-4
 Distance "FROM" to "TO" 300 cm
 Inside Diameter 2900 mm
 Element Thickness 183 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 183 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 194.76 kgf/cm²
 Design Temperature Internal Pressure 290 °C
 Design External Pressure 0 kgf/cm²
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-387 11 +IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 60
 Detail Type Packing
 Detail ID Bed-3 4
 Dist. from "FROM" Node / Offset dist 0 cm
 Height of Packed Section 137.5 cm
 Density 0.0023 kg/cm³
 Percent Volume Holdup 0.0
 Specific Gravity of Packing Liquid 0.0

Element From Node 60
 Detail Type Packing
 Detail ID Bed-2 1
 Dist. from "FROM" Node / Offset dist 200 cm
 Height of Packed Section 100 cm
 Density 0.0013 kg/cm³
 Percent Volume Holdup 0.0
 Specific Gravity of Packing Liquid 0.0

Element From Node 60
 Detail Type Packing
 Detail ID Basket
 Dist. from "FROM" Node / Offset dist 0 cm
 Height of Packed Section 300 cm
 Density 0.000649 kg/cm³
 Percent Volume Holdup 0.0
 Specific Gravity of Packing Liquid 0.0

Element From Node 60
 Detail Type Insulation
 Detail ID Ins: 60
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 300 cm
 Thickness of Insulation 100 mm
 Density 0.0002 kg/cm³

Element# 7/13
 Element From Node 70
 Element To Node 80
 Element Type Cylinder
 Description Shell-5
 Distance "FROM" to "TO" 300 cm
 Inside Diameter 2900 mm
 Element Thickness 183 mm
 Internal Corrosion Allowance 1.5 mm



Nominal Thickness	183	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	194.76	kgf/cm ²
Design Temperature Internal Pressure	290	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Packing	
Detail ID	Bed-2 2	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	300	cm
Density	0.0013	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	70	
Detail Type	Packing	
Detail ID	Basket	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	300	cm
Density	0.000649	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	100	mm
Density	0.0002	kg/cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	T8 2"	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.906	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

Element From Node	70	
Detail Type	Nozzle	
Detail ID	T9 2"	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.906	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

Element From Node	70	
Detail Type	Nozzle	
Detail ID	T10 2"	
Dist. from "FROM" Node / Offset dist	250	cm
Nozzle Diameter	50	mm
Nozzle Schedule	None	



Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.906	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

Element#	8/13	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	300	cm
Inside Diameter	2900	mm
Element Thickness	183	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	183	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	194.76	kgf/cm ²
Design Temperature Internal Pressure	290	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Packing	
Detail ID	Bed-2 3	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	300	cm
Density	0.0013	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	80	
Detail Type	Packing	
Detail ID	Basket	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	300	cm
Density	0.000649	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 80	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	100	mm
Density	0.0002	kg/cm ³

Element From Node	80	
Detail Type	Weight	
Detail ID	Cleat wt	
Dist. from "FROM" Node / Offset dist	200	cm
Miscellaneous Weight	2650	kgf
Offset from Element Centerline	0	mm

Element From Node	80	
Detail Type	Weight	
Detail ID	10MT MISC wt	
Dist. from "FROM" Node / Offset dist	250	cm
Miscellaneous Weight	9999.8	kgf
Offset from Element Centerline	0	mm



Element#	9/13
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	Shell-7
Distance "FROM" to "TO"	300 cm
Inside Diameter	2900 mm
Element Thickness	183 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	183 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.76 kgf/cm ²
Design Temperature Internal Pressure	290 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Packing
Detail ID	Bed-2 4
Dist. from "FROM" Node / Offset dist	0 cm
Height of Packed Section	233.5 cm
Density	0.0013 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	90
Detail Type	Packing
Detail ID	Bed-1 1
Dist. from "FROM" Node / Offset dist	250 cm
Height of Packed Section	50 cm
Density	0.00095 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	90
Detail Type	Packing
Detail ID	Basket
Dist. from "FROM" Node / Offset dist	0 cm
Height of Packed Section	300 cm
Density	0.000649 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 90
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	300 cm
Thickness of Insulation	100 mm
Density	0.0002 kg/cm ³

Element#	10/13
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Shell-8
Distance "FROM" to "TO"	300 cm
Inside Diameter	2900 mm
Element Thickness	183 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	183 mm



External Corrosion Allowance	0	mm
Design Internal Pressure	194.76	kgf/cm ²
Design Temperature Internal Pressure	290	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Packing	
Detail ID	Bed-1 2	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	300	cm
Density	0.00095	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	100	
Detail Type	Packing	
Detail ID	Basket	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	300	cm
Density	0.000649	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	100	
Detail Type	Insulation	
Detail ID	Ins: 100	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	300	cm
Thickness of Insulation	100	mm
Density	0.0002	kg/cm ³

Element#	11/13	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-9	
Distance "FROM" to "TO"	280.18	cm
Inside Diameter	2900	mm
Element Thickness	183	mm
Internal Corrosion Allowance	1.5	mm
Nominal Thickness	183	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	194.76	kgf/cm ²
Design Temperature Internal Pressure	290	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	110	
Detail Type	Packing	
Detail ID	Bed-1 3	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	252	cm
Density	0.00095	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	110	
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Detail Type	Packing	
Detail ID	Basket	
Dist. from "FROM" Node / Offset dist	0	cm
Height of Packed Section	280.18	cm
Density	0.000649	kg/cm ³
Percent Volume Holdup	0.0	
Specific Gravity of Packing Liquid	0.0	

Element From Node	110	
Detail Type	Insulation	
Detail ID	Ins: 110	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	280.18	cm
Thickness of Insulation	100	mm
Density	0.0002	kg/cm ³

Element From Node	110	
Detail Type	Nozzle	
Detail ID	T5 2"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.906	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11 +IT	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	T6 2"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.906	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11 +IT	

Element From Node	110	
Detail Type	Nozzle	
Detail ID	T7 2"	
Dist. from "FROM" Node / Offset dist	200	cm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.906	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11 +IT	

Element From Node	110	
Detail Type	Weight	
Detail ID	Trunnion-1	
Dist. from "FROM" Node / Offset dist	180	cm
Miscellaneous Weight	749.99	kgf
Offset from Element Centerline	1500	mm

Element From Node	110	
Detail Type	Weight	
Detail ID	Trunnion-2	
Dist. from "FROM" Node / Offset dist	180	cm
Miscellaneous Weight	749.99	kgf
Offset from Element Centerline	1500	mm

Element From Node	110	
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Detail Type	Weight
Detail ID	A4 nozzle wt
Dist. from "FROM" Node / Offset dist	270 cm
Miscellaneous Weight	2000 kgf
Offset from Element Centerline	1500 mm

Element#	12/13
Element From Node	120
Element To Node	130
Element Type	Flange
Description	Shell flange
Distance "FROM" to "TO"	127.5 cm
Flange Inside Diameter	2900 mm
Element Thickness	1000 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	1000 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.76 kgf/cm ²
Design Temperature Internal Pressure	290 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11 +IT
Allowable Stress, Ambient	2110.7 kgf/cm ²
Allowable Stress, Operating	1680.4 kgf/cm ²
Allowable Stress, Hydrotest	3002.9 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2520.6 kgf/cm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-3
UNS Number	K11572
Class / Thickness / Grade	3 ::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	13/13
Element From Node	130
Element To Node	140
Element Type	Flange
Description	Top cover flang
Distance "FROM" to "TO"	45 cm
Flange Outside Diameter	4000 mm
Element Thickness	700 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	700 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	194.76 kgf/cm ²
Design Temperature Internal Pressure	290 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-336 F11 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	130
Detail Type	Nozzle
Detail ID	T1 2"
Dist. from "FROM" Node / Offset dist	1300 mm
Nozzle Diameter	50 mm
Nozzle Schedule	None
Nozzle Class	1500
Layout Angle	0.0
Blind Flange (Y/N)	N



Weight of Nozzle (Used if > 0)	23.773	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT
Element From Node	130	
Detail Type	Nozzle	
Detail ID	T2 2"	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	23.773	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT
Element From Node	130	
Detail Type	Nozzle	
Detail ID	T3 2"	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	23.773	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT
Element From Node	130	
Detail Type	Nozzle	
Detail ID	T4 2"	
Dist. from "FROM" Node / Offset dist	1300	mm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	1500	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	23.773	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-336 F11	+IT

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
Bottom head		...	...	...	...
Shell-1		...	435	...	435
Skirt		...	435	...	-300
Shell-2		...	600	...	165
Shell-3		...	900	...	300
Shell-4		...	1200	...	300
Shell-5		...	1500	...	300
Shell-6		...	1800	...	300
Shell-7		...	2100	...	300
Shell-8		...	2400	...	300
Shell-9		...	2680.18	...	280.18
Shell flange		...	2808	...	127.5
Top cover flang		...	2853	...	45

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

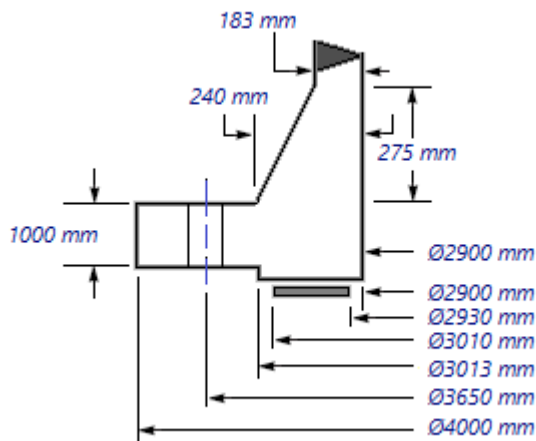
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Flange Input Data Values

Description: Shell flange :

Shell flange

Flange Type	Integral Weld Neck		
Design Pressure	P	194.76	kgf/cm ²
Design Temperature		290	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	2900.000	mm
Flange Outside Diameter	A	4000.000	mm
Flange Thickness	t	1000.0000	mm
Thickness of Hub at Small End	go	183.0000	mm
Thickness of Hub at Large End	gl	240.0000	mm
Length of Hub	h	275.0000	mm
Flange Material	SA-336 F11 +IT		
Flange Material UNS number	K11572		
Flange Allowable Stress At Temperature	Sfo	1680.42	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	2110.70	kgf/cm ²
Bolt Material	SA-540 B23		
Bolt Allowable Stress At Temperature	Sb	2039.32	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	2039.32	kgf/cm ²
Diameter of Bolt Circle	C	3650.000	mm
Nominal Bolt Diameter	a	150.0000	mm
Type of Threads	Special, Root Area		
Root Area of a Single Bolt		172.6500	cm ²
Number of Bolts		40	
Flange Face Outside Diameter	Fod	3013.000	mm
Flange Face Inside Diameter	Fid	2900.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	3010.000	mm
Gasket Inside Diameter	Gi	2930.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P(R + Ca) / (S - 0.5 \cdot P)) + (Ca + Ca_{ext})$$



$$= (194.759(2900.0+1.5)/(1680.424-0.5 \cdot 194.759)) + (1.5)$$

$$= 180.0754 \text{ mm}$$

Small End Hub MAWP:

$$= S \cdot \log((2 \cdot T + R) / R)$$

$$= 1680.424 \cdot \log((2 \cdot 181.5 + 1451.5) / 1451.5)$$

$$= 197.990 \text{ kgf/cm}^2$$

Small End Hub MAPnc:

$$= S_a \cdot \log((2.0 \cdot T + R) / R)$$

$$= 2110.699 \cdot \log((2.0 \cdot 183.0 + 1450.0) / 1450.0)$$

$$= 250.868 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	2903.000 mm
Corroded Large Hub,	g1Cor = g1-ci	238.500 mm
Corroded Small Hub,	g0Cor = go-ci	181.500 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	135.000 mm

Gasket Contact Width,	N = Self-Energizing	0.000 mm
Basic Gasket Width,	bo = N/2	0.000 mm
Effective Gasket Width,	b = bo	0.000 mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	3010.000 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 3010.0^2 \cdot 194.759$$

$$= 13858618.000 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0.0 \cdot \pi \cdot 3010.0 \cdot 0.0 \cdot 194.76$$

$$= 0.000 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 2903.0^2 \cdot 194.7591$$

$$= 12890835.000 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 13858618 - 12890835$$

$$= 967783.000 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(13858618 + 0 + 0, 0)$$

$$= 13858618.000 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 0.0 \cdot 0.0 \cdot \pi \cdot 3010.0 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 0.000 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((13858618 + 0 + 4 \cdot 0 / 3010.0) / 2039, 0 / 2039)$$

$$= 6795.699 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 150.0 + 6 \cdot 1000.0 / (0.0 + 0.5)$$

$$= 12300.000 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 3650.0 \cdot \sin(\pi / 40)$$

$$= 286.376 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{286.376}{(2 \cdot 150.0 + 1000.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for User Defined Bolts (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6795.699	6905.999	
Circ. Spacing between the Bolts:	0.000	286.376	12300.000

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 2039.32 \cdot 6906.0$$

$$= 14083555.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 13858618 - 13858618$$

$$= 0.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (3650.0 - 3010.0) / 2$$

$$= 320.0000 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 \cdot (C - B) / 2 + hg$$

$$= 0.5 \cdot (3650.0 - 2903.0) / 2 + 320.0$$

$$= 346.7500 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (3650.0 - 2903.0 - 238.5) / 2$$

$$= 254.2500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	12890835.	254.2500	1.0000	3277498.
Face Pressure, Mt	967783.	346.7500	1.0000	335579.
Gasket Load, Mg	0.	320.0000	1.0000	0.
Gasket Seating, Matm	14083555.	320.0000	1.0000	4506744.

Total Moment for Operation, Mo 3613078. kgf-m.
 Total Moment for Gasket seating, Mg 4506744. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 725.875 mm
 Hub Ratio, h/h0 = HL / H0 0.379
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.314

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.878
 Factor V per 4.16.5 0.395
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 4000.0 / 2903.0$$

$$= 1.378$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$$

$$= (0.66845 + (5.7169 \cdot 1.378^2 \cdot \log_{10}(1.378))) / (1.378^2 - 1) / (1.378 - 1)$$

$$= 6.219$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.378^2(1 + 8.55246 \cdot \log_{10}(1.378)) - 1) / ((1.0472 + 1.9448 \cdot 1.378^2) \cdot (1.378 - 1))$$

$$= 1.764$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.378^2(1 + 8.55246 \cdot \log_{10}(1.378)) - 1) / (1.36136(1.378^2 - 1) \cdot (1.378 - 1))$$

$$= 6.834$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.378^2 + 1) / (1.378^2 - 1)$$

$$= 3.226$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 6.834 \cdot 181.5^2 \cdot 725.875 / 0.395$$

$$= 413643488.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.878 / 725.875$$

$$= 0.001 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (1000.0 \cdot 0.001 + 1) / 1.764 + 1000.0^3 / 41364E+09$$

$$= 3.670$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 3613078 / 2903.0) / (3.6702 \cdot 238.5^2)$$

$$= 596.16 \text{ kgf/cm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.0 \cdot 4506744 / 2903.0) / (3.6702 \cdot 238.5^2)$$

$$= 743.61 \text{ kgf/cm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.6127 \cdot 3613078 / 2903.0) / (3.6702 \cdot 1000.0^2)$$

$$= 88.60 \text{ kgf/cm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.6127 \cdot 4506744 / 2903.0) / (3.6702 \cdot 1000.0^2)$$

$$= 110.51 \text{ kgf/cm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (6.2189 \cdot 3613078 / (1000.0^2 \cdot 2903.0)) - 3.2258 \cdot 89$$

$$= 488.20 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (6.2189 \cdot 4506744 / (1000.0^2 \cdot 2903.0)) - 3.2258 \cdot 111$$

$$= 608.96 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (596 + \max(89, 488)) / 2$$

$$= 542.18 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (744 + \max(111, 609)) / 2$$

$$= 676.28 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BS_o]:

$$= W_{m1} / A_b$$

$$= 13858618 / 6906.0$$

$$= 2006.75 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BS_a]:

$$= (W_{m2} / A_b)$$

$$= (0 / 6906.0)$$

$$= 0.00 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	596.16	2520.64	743.61 3166.05
Radial Flange	88.60	1680.42	110.51 2110.70
Tangential Flange	488.20	1680.42	608.96 2110.70
Maximum Average	542.18	1680.42	676.28 2110.70
Bolting	2006.75	2039.32	0.00 2039.32

Minimum Required Flange Thickness 410.743 mm

Estimated M.A.W.P. (Operating) 198 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 50607.7 kg
 Estimated Unfinished Weight of Forging at given Thk 58906.6 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [J_g]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 4506744 / 1.0 \cdot 100000.023 \cdot 0.395 / (1.016 \cdot 2081087 \cdot 181.5^2 \cdot 725.875 \cdot 0.3)$$

$$= 0.611 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [J_o]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 3613078 / 1.0 \cdot 100000.023 \cdot 0.395 / (1.016 \cdot 1907444 \cdot 181.5^2 \cdot 725.875 \cdot 0.3)$$

$$= 0.535 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [J_g]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 4506744 / 1.0 \cdot 100000.023 \cdot 0.395 / (3.67 \cdot 2081087 \cdot 181.5^2 \cdot 725.875 \cdot 0.3)$$

$$= 0.169 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [J_o]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_{oc}^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 3613078 / 1.0 \cdot 100000.023 \cdot 0.395 / (3.67 \cdot 1907444 \cdot 181.5^2 \cdot 725.875 \cdot 0.3)$$

$$= 0.148 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	597.21	3166.05	742.28 3166.05
Radial Flange	89.70	2110.70	111.49 2110.70
Tangential Flange	486.67	2110.70	604.89 2110.70
Maximum Average	541.94	2110.70	673.59 2110.70



Bolting	2006.75	2039.32	0.00	2039.32
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Minimum Req'd Thickness, ambient, no Corrosion: 407.264 mm

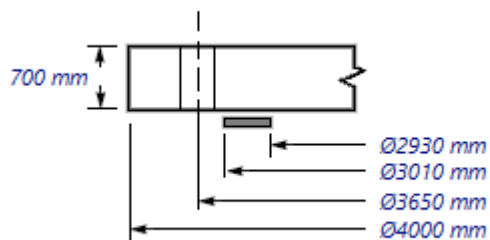
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Flange Input Data Values

Description: Topcover flg :

Top cover flang

Flange Type		Blind	
Design Pressure	P	194.76	kgf/cm ²
Design Temperature		290	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	4000.000	mm
Flange Thickness	t	700.0000	mm
Flange Material		SA-336 F11 +IT	
Flange Material UNS number		K11572	
Flange Allowable Stress At Temperature	Sfo	1680.42	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	2110.70	kgf/cm ²
Bolt Material		SA-540 B23	
Bolt Allowable Stress At Temperature	Sb	2039.32	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	2039.32	kgf/cm ²
Perimeter along the Center of the Bolts	L	11466.813	mm
Diameter of Bolt Circle	C	3650.000	mm
Nominal Bolt Diameter	a	150.0000	mm
Type of Threads	Special,	Root Area	
Root Area of a Single Bolt		172.6500	cm ²
Number of Bolts		40	
Flange Face Outside Diameter	Fod	3013.000	mm
Flange Face Inside Diameter	Fid	2900.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	3010.000	mm
Gasket Inside Diameter	Gi	2930.000	mm
Gasket Factor	m	0.0000	
Gasket Design Seating Stress	y	0.00	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	N = Self-Energizing	0.000 mm
Basic Gasket Width,	bo = N/2	0.000 mm
Effective Gasket Width,	b = bo	0.000 mm
Gasket Reaction Diameter,	G = Go (Self-Energizing)	3010.000 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 3010.0^2 \cdot 194.759$
 $= 13858618.000 \text{ kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 0.0 \cdot \pi \cdot 3010.0 \cdot 0.0 \cdot 194.76$$

$$= 0.000 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(13858618 + 0 + 0, 0)$$

$$= 13858618.000 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 0.0 \cdot 0.0 \cdot \pi \cdot 3010.0 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 0.000 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((13858618 + 0 + 4 \cdot 0/3010.0)/2039, 0/2039)$$

$$= 6795.699 \text{ cm}^2$$

Bolting Information for User Defined Bolts (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	6795.699	6906.001	
Circ. Spacing between the Bolts:	...	286.376	...

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 2039.32 \cdot 6906.001$$

$$= 14083559.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 13858618.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (3650.0 - 3010.0) / 2$$

$$= 320.0000 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 13858618 \cdot 320.0 \cdot 1.0 / (698.5^2 \cdot 3010.0) + 0.3 \cdot 1.0 \cdot 194.76 \cdot 3010.0^2 / 698.5^2$$

$$= 1658.73 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 14083559 \cdot 320.0 \cdot 1.0 / (698.5^2 \cdot 3010.0)$$

$$= 583.07 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 13858618 / 6906.001$$

$$= 2006.75 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (0 / 6906.001)$$

$$= 0.00 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

Operating

Gasket Seating



	Actual	Allowed	Actual	Allowed
Tangential Flange	1658.73	1680.42	583.07	2110.70
Bolting	2006.75	2039.32	0.00	2039.32

Reqd. Blind Flange Thickness at Center 695.476 mm
 Reqd. Blind Flange Thickness at Gasket 409.650 mm

Estimated M.A.W.P. (Operating) 197 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 68175.8 kg
 Estimated Unfinished Weight of Forging at given Thk 68175.8 kg

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Tangential Flange	1651.63	2110.70	580.57	2110.70
Bolting	2006.75	2039.32	0.00	2039.32

Minimum Req'd Thickness, ambient, no Corrosion: 619.227 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
Bottom head		194.76	107	1.5	2992	1680.4	1
Shell-1		194.76	183	3	2900	1680.4	2
Skirt		...	60	3	3265.7	...	3
Shell-2		194.76	183	1.5	2900	1680.4	4
Shell-3		194.76	183	1.5	2900	1680.4	5
Shell-4		194.76	183	1.5	2900	1680.4	6
Shell-5		194.76	183	1.5	2900	1680.4	7
Shell-6		194.76	183	1.5	2900	1680.4	8
Shell-7		194.76	183	1.5	2900	1680.4	9
Shell-8		194.76	183	1.5	2900	1680.4	10
Shell-9		194.76	183	1.5	2900	1680.4	11
Shell flange		194.76	1000	1.5	2900	1680.4	12
Top cover flang		194.76	700	1.5	4000	1680.4	13

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
Bottom head		194.759	199.325	254.593	93	90.8429	1
Shell-1		194.759	196.254	250.868	183	181.548	2
Skirt		...	No Calc	No Calc	60	No Calc	3
Shell-2		194.759	197.99	250.868	183	179.864	4
Shell-3		194.759	197.99	250.868	183	179.864	5
Shell-4		194.759	197.99	250.868	183	179.864	6
Shell-5		194.759	197.99	250.868	183	179.864	7
Shell-6		194.759	197.99	250.868	183	179.864	8
Shell-7		194.759	197.99	250.868	183	179.864	9
Shell-8		194.759	197.99	250.868	183	179.864	10
Shell-9		194.759	197.99	250.868	183	179.864	11
Shell flange		194.759	197.92	197.92	1000	410.743	12
Top cover flang		194.759	197.306	197.92	700	695.476	13

Minimum

196

198

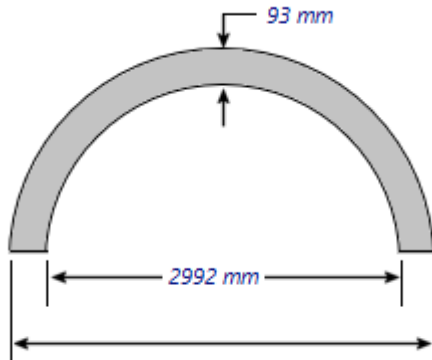
MAWP: 196 kgf/cm², limited by: Shell-1.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Bottom head

Spherical Head Design Information:					
Design Pressure	P	194.759 kgf/cm ²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		0.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm ²
Inside Diameter	Di	2992.000 mm	Actual Stress	1546.5 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		93.0000 mm	Spec. Nominal Thk.	107.000 mm	
Surface Area		161453.3 cm ²	SG of Contents	0.000	
Element Volume		7012202.5 cm ³	Weight of Contents	0.0 kgf	
Empty Weight		12515.3 kgf	Operating Weight	12515.3 kgf	
			Comp. All. Ope. Stress	948.5 kgf/cm ²	
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.400 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.700 mm

Material treatment options: +IT

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:
 $= D/2(\exp(0.5P/(S \cdot E)) - 1) + c_i + c_o$
 $= 2995.0/2(\exp(0.5 \cdot 194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 89.3429 + 1.5 + 0.0 = 90.8429 \text{ mm}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 2 \cdot 1680.4 \cdot 1.0 \cdot \ln((2 \cdot 91.5 + 2995.0) / 2995.0) - 0.0$
 $= 199.325 - 0.0 = 199.3249 \text{ kgf/cm}^2$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:
 $= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D) / D)$
 $= 2 \cdot 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 93.0 + 2992.0) / 2992.0)$
 $= 254.5928 \text{ kgf/cm}^2$



Computed Stress Intensity at Design Pressure [Si]:

$$\begin{aligned} &= P \cdot D^2 / (E (D_o^2 - D^2)) \\ &= 194.759 \cdot 2995.0^2 / (1.0 (3178.0^2 - 2995.0^2)) \\ &= 1546.477 \text{ kgf/cm}^2 \end{aligned}$$

Computed Membrane Stress at Design Pressure [Sm]:

$$\begin{aligned} &= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D)) \\ &= 0.5 \cdot 194.759 / (1.0 \cdot \ln((2 \cdot 91.5 + 2995.0) / 2995.0)) \\ &= 1641.932 \text{ kgf/cm}^2 \end{aligned}$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$	5.179 %
For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$	41.566 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 161453.3 / 3.14159) \\ &= 4533.969 \text{ mm} \end{aligned}$$

MDMT Calculations in the Spherical Portion:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

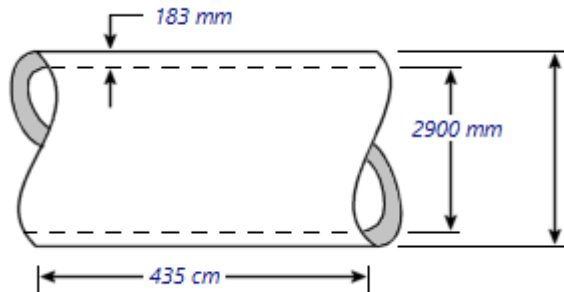
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.400 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.700 mm

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		435.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1572.1 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		446329.3 cm²	SG of Contents	0.000	
Element Volume		28732686.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		59756.6 kgf	Operating Weight	59756.6 kgf	
			Comp. All. Ope. Stress	1038.7 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2906.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 178.5479 + 3.0 + 0.0 = 181.5479 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 180.0 + 2906.0) / 2906.0) - 0.0$$

$$= 196.254 - 0.0 = 196.254 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8676 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2906.0 / (1.0(3266.0 - 2906.0))$$

$$= 1572.1390 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 180.0 + 2906.0) / 2906.0))$$
$$= 1667.6235 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

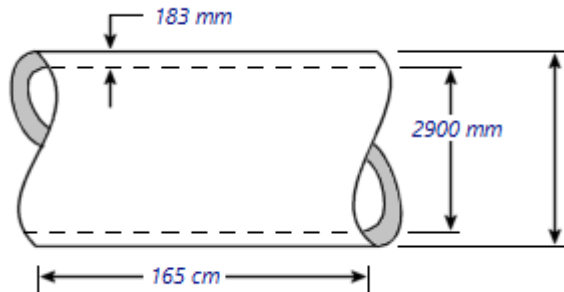
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		165.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		169297.3 cm²	SG of Contents	0.000	
Element Volume		10898606.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		22666.3 kgf	Operating Weight	22666.3 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9897 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8676 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5364 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0040 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

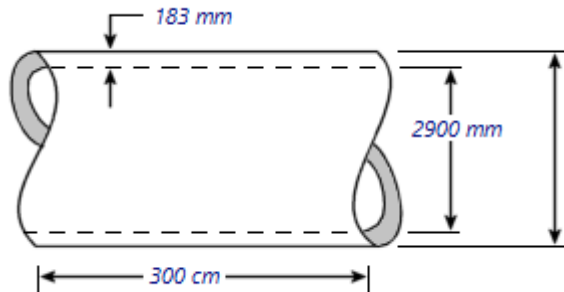
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		307813.3 cm²	SG of Contents	0.000	
Element Volume		19815642.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		41211.5 kgf	Operating Weight	41211.5 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

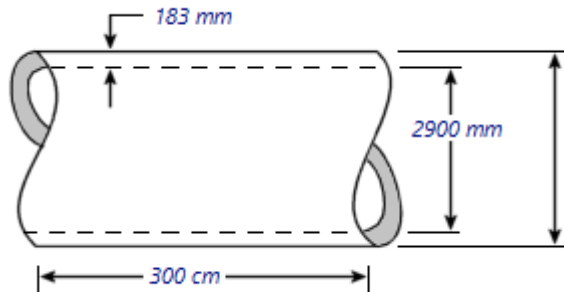
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or sa		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		307813.3 cm²	SG of Contents	0.000	
Element Volume		19815642.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		41211.5 kgf	Operating Weight	41211.5 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

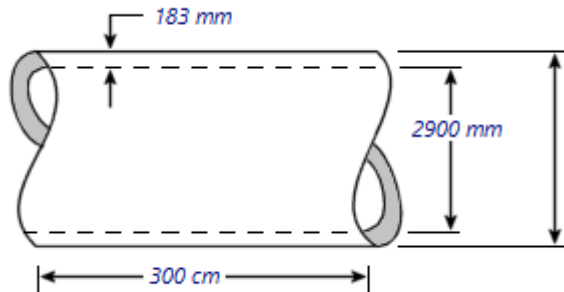
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		307813.3 cm²	SG of Contents	0.000	
Element Volume		19815642.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		41211.5 kgf	Operating Weight	41211.5 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

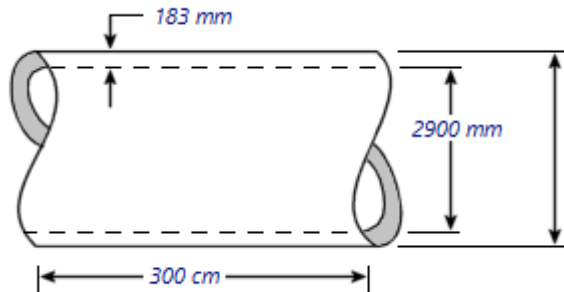
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		307813.3 cm²	SG of Contents	0.000	
Element Volume		19815642.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		41211.5 kgf	Operating Weight	41211.5 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

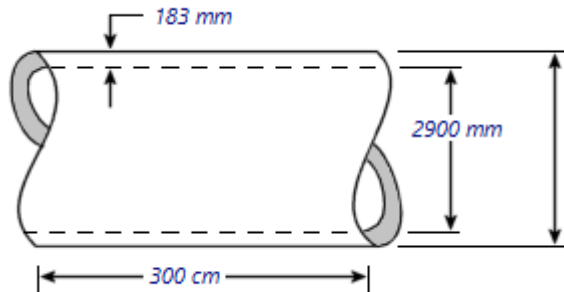
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope) S	1680.4 kgf/cm²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	2110.7 kgf/cm²	
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		307813.3 cm²	SG of Contents	0.000	
Element Volume		19815642.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		41211.5 kgf	Operating Weight	41211.5 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

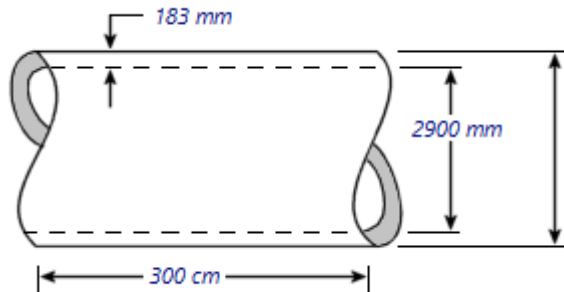
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-8

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		300.000 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		307813.3 cm²	SG of Contents	0.000	
Element Volume		19815642.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		41211.5 kgf	Operating Weight	41211.5 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

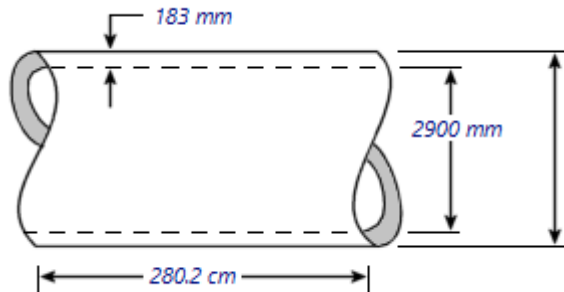
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-9

Cylindrical Shell Design Information:					
Design Pressure	P	194.759 kgf/cm²	Int. Design Temperature	290 °C	
Material		SA-387 11	Ext. Design Temperature	0 °C	
Length		280.180 cm	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1680.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	2110.7 kgf/cm²
Inside Diameter	Di	2900.000 mm	Actual Stress	1557.5 kgf/cm²	
Spec. Min. Thk t _e , or s _a		183.0000 mm	Spec. Nominal Thk.	183.000 mm	
Surface Area		287477.1 cm²	SG of Contents	0.000	
Element Volume		18506488.0 cm³	Weight of Contents	0.0 kgf	
Empty Weight		38488.8 kgf	Operating Weight	38488.8 kgf	
			Comp. All. Ope. Stress	1039.8 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2903.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 178.3636 + 1.5 + 0.0 = 179.8636 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0) - 0.0$$

$$= 197.99 - 0.0 = 197.9899 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 2110.699 \cdot 1.0 \cdot \ln((2 \cdot 183.0 + 2900.0) / 2900.0)$$

$$= 250.8678 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E(D_o - D))$$

$$= 194.759 \cdot 2903.0 / (1.0(3266.0 - 2903.0))$$

$$= 1557.5356 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 194.759 / (1.0 \cdot \ln((2 \cdot 181.5 + 2903.0) / 2903.0))$$
$$= 1653.0024 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.936 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

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Hydrostatic Test Pressure Results:

User Defined Hydrostatic Test Pressure at High Point 310.590 kgf/cm²

Horizontal Test performed per: User Hydro Pressure

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	2577.4	3002.9	0.858	310.89
Shell-1	2615.6	3002.9	0.871	310.88
Shell-2	2615.6	3002.9	0.871	310.88
Shell-3	2615.6	3002.9	0.871	310.88
Shell-4	2615.6	3002.9	0.871	310.88
Shell-5	2615.6	3002.9	0.871	310.88
Shell-6	2615.6	3002.9	0.871	310.88
Shell-7	2615.6	3002.9	0.871	310.88
Shell-8	2615.6	3002.9	0.871	310.88
Shell-9	2615.6	3002.9	0.871	310.88

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A2 12"	Nozzle	2110.70	1680.42	1.256
A1 12"	Nozzle	2110.70	1680.42	1.256
T11 2"	Nozzle	2110.70	1680.42	1.256
T12 2"	Nozzle	2110.70	1680.42	1.256
T13 2"	Nozzle	2110.70	1680.42	1.256
T8 2"	Nozzle	2110.70	1680.42	1.256
T9 2"	Nozzle	2110.70	1680.42	1.256
T10 2"	Nozzle	2110.70	1680.42	1.256
T5 2"	Nozzle	2110.70	1680.42	1.256
T6 2"	Nozzle	2110.70	1680.42	1.256
T7 2"	Nozzle	2110.70	1680.42	1.256
T1 2"	Nozzle	2110.70	1680.42	1.256
T2 2"	Nozzle	2110.70	1680.42	1.256
T3 2"	Nozzle	2110.70	1680.42	1.256
T4 2"	Nozzle	2110.70	1680.42	1.256

Minimum 1.256

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
Bottom head	2110.70	1680.42	1.256
Shell-1	2110.70	1680.42	1.256
Shell-2	2110.70	1680.42	1.256
Shell-3	2110.70	1680.42	1.256
Shell-4	2110.70	1680.42	1.256
Shell-5	2110.70	1680.42	1.256
Shell-6	2110.70	1680.42	1.256
Shell-7	2110.70	1680.42	1.256
Shell-8	2110.70	1680.42	1.256
Shell-9	2110.70	1680.42	1.256
Shell flange	2110.70	1680.42	1.256
Top cover flang	2110.70	1680.42	1.256

Minimum 1.256

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):



Description	Ambient	Operating	Ratio
A2 12"	2271.60	3002.91	0.756
A1 12"	2271.60	3002.91	0.756
T11 2"	1119.76	3002.91	0.373
T12 2"	1119.76	3002.91	0.373
T13 2"	1119.76	3002.91	0.373
T8 2"	1119.76	3002.91	0.373
T9 2"	1119.76	3002.91	0.373
T10 2"	1119.76	3002.91	0.373
T5 2"	1119.76	3002.91	0.373
T6 2"	1119.76	3002.91	0.373
T7 2"	1119.76	3002.91	0.373
T1 2"	1120.15	3002.91	0.373
T2 2"	1120.15	3002.91	0.373
T3 2"	1120.15	3002.91	0.373
T4 2"	1120.15	3002.91	0.373

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	3178	91.5	No Calc	No Calc
20	30	2730.05	3266	180	No Calc	No Calc
30	40	No Calc	...	...	No Calc	No Calc
40	50	2730.05	3266	181.5	No Calc	No Calc
50	60	2730.05	3266	181.5	No Calc	No Calc
60	70	2730.05	3266	181.5	No Calc	No Calc
70	80	2730.05	3266	181.5	No Calc	No Calc
80	90	2730.05	3266	181.5	No Calc	No Calc
90	100	2730.05	3266	181.5	No Calc	No Calc
100	110	2730.05	3266	181.5	No Calc	No Calc
110	120	2730.05	3266	181.5	No Calc	No Calc
120	130	No Calc	...	998.5	No Calc	No Calc
130	140	No Calc	...	698.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	93	No Calc	...	166.141
20	30	183	No Calc	...	143.624
30	40	...	No Calc	...	No Calc
40	50	183	No Calc	...	145.385
50	60	183	No Calc	...	145.385
60	70	183	No Calc	...	145.385
70	80	183	No Calc	...	145.385
80	90	183	No Calc	...	145.385
90	100	183	No Calc	...	145.385
100	110	183	No Calc	...	145.385
110	120	183	No Calc	...	145.385
120	130	1000	408.635	...	No Calc
130	140	700	368.623	...	No Calc

Minimum

144

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2730.05	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	2730.05	No Calc	No Calc	No Calc
50	60	2730.05	No Calc	No Calc	No Calc
60	70	2730.05	No Calc	No Calc	No Calc
70	80	2730.05	No Calc	No Calc	No Calc
80	90	2730.05	No Calc	No Calc	No Calc
90	100	2730.05	No Calc	No Calc	No Calc
100	110	2730.05	No Calc	No Calc	No Calc
110	120	2730.05	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc
130	140	No Calc	No Calc	No Calc	No Calc



Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	12515.3	7012202	12351.7	7033316	...
20	30	59756.6	28732686	58834.2	28851704	...
30	40	18076.6	...	17688.5	...	...
40	50	22666.3	10898606	22491.5	10921165	...
50	60	41211.5	19815642	40893.6	19856664	...
60	70	41211.5	19815642	40893.6	19856664	...
70	80	41211.5	19815642	40893.6	19856664	...
80	90	41211.5	19815642	40893.6	19856664	...
90	100	41211.5	19815642	40893.6	19856664	...
100	110	41211.5	19815642	40893.6	19856664	...
110	120	38488.8	18506488	38191.9	18544800	...
120	130	50607.7	8435321	50472.6	8439082	...
130	140	68175.8	...	68175.8	...	...
Total		517556	0.192E+09	513567	0.193E+09	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Nozl	255.117	100	-131.616	...	A2 12"
10	Nozl	255.117	-100	-131.616	...	A1 12"
10	Wght	19999.6	...	-140	...	Bottom assembly
20	Pack	62709.3	...	217.5	...	Bed-3 1
20	Pack	18669	...	217.5	...	Basket
20	Insl	275.257	...	65	...	IN
20	Wght	3749.93	150	...	...	SF weight
20	Wght	7000	-150	...	...	Sup ring wt
40	Pack	23786.3	...	82.5	...	Bed-3 2
40	Pack	7081.34	...	82.5	...	Basket
40	Insl	349.364	...	82.5	...	INS
40	Nozl	50.9063	160	20	...	T11 2"
40	Nozl	50.9063	113.137	20	-113.137	T12 2"
40	Nozl	50.9063	...	20	-160	T13 2"
40	Wght	25599.5	...	100	...	Skirt wt
50	Pack	43247.8	...	150	...	Bed-3 3
50	Pack	12875.2	...	150	...	Basket
50	Insl	635.207	...	150	...	Ins: 50
60	Pack	20913	...	68.75	...	Bed-3 4
60	Pack	8596.66	...	250	...	Bed-2 1
60	Pack	12875.2	...	150	...	Basket
60	Insl	635.207	...	150	...	Ins: 60
70	Pack	25790	...	150	...	Bed-2 2
70	Pack	12875.2	...	150	...	Basket
70	Insl	635.207	...	150	...	Ins: 70
70	Nozl	50.9063	160	250	...	T8 2"
70	Nozl	50.9063	113.137	250	-113.137	T9 2"
70	Nozl	50.9063	...	250	-160	T10 2"
80	Pack	25790	...	150	...	Bed-2 3
80	Pack	12875.2	...	150	...	Basket
80	Insl	635.207	...	150	...	Ins: 80
80	Wght	2649.95	...	200	...	Cleat wt
80	Wght	9999.82	...	250	...	10MT Misc wt
90	Pack	20073.2	...	116.75	...	Bed-2 4
90	Pack	3141.09	...	275	...	Bed-1 1
90	Pack	12875.2	...	150	...	Basket
90	Insl	635.207	...	150	...	Ins: 90
100	Pack	18846.5	...	150	...	Bed-1 2
100	Pack	12875.2	...	150	...	Basket
100	Insl	635.207	...	150	...	Ins: 100
110	Pack	15831.1	...	126	...	Bed-1 3



110	Pack	12024.5	...	140.09	...	Basket
110	Insl	593.241	...	140.09	...	Ins: 110
110	Nozl	50.9063	160	200	...	T5 2"
110	Nozl	50.9063	113.137	200	-113.137	T6 2"
110	Nozl	50.9063	...	200	-160	T7 2"
110	Wght	749.987	150	180	...	Trunnion-1
110	Wght	749.987	-150	180	...	Trunnion-2
110	Wght	1999.96	150	270	...	A4 nozzle wt
130	Nozl	23.7733	130	60	...	T1 2"
130	Nozl	23.7733	...	60	-130	T2 2"
130	Nozl	23.7733	-130	60	...	T3 2"
130	Nozl	23.7733	...	60	130	T4 2"

Total Weight of Each Detail Type:

Packing	383750.8
Insulation	5029.1
Nozzles	1063.5
Weights	72498.8
Sum of the Detail Weights	462342.2 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	517556.0	517556.0	517556.0	517556.0	517556.0	517556.0
Nozzles	1063.5	1063.5	1063.5	1063.5	1063.5	1063.5
Wld Weights	72498.8	72498.8	72498.8	72498.8	72498.8	72498.8
Packing	...	...	383750.8	383750.8	383750.8	383750.8
Insulation	...	...	5029.1	5029.1	5029.1	5029.1
Test Liquid	...	192392.3	...	...	...	...
Totals	591118.3	783510.7	979898.2	979898.2	979898.2	979898.2

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	591118.3 kg
Shop Test Wt.	- Fabricated weight + water (full)	783510.7 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	979898.2 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	979898.2 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	979898.2 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	979898.2 kg

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	312640.0 kg
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	346234.2 kg
Mass of the Upper 1/3 of the Vertical Vessel (empty)	313599.8 kg

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	161453	140619
20	30	446329	396312
30	40	319098	307788
40	50	169297	150325
50	60	307813	273319
60	70	307813	273319
70	80	307813	273319
80	90	307813	273319
90	100	307813	273319
100	110	307813	273319
110	120	287477	255261



120	130	214477	116160
130	140	213628	...

Total 3658639.5 3006377.5 cm²
 Total 3938.1 3236.0 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg	Total. Ele. Oper. Wgt. kg	Total. Ele. Hydro. Wgt. kg	Total Dtl. Offset Mom. kgf-m.	Oper. Wgt. No Liquid kg
10	20	33025.1	33025.1	40034.2	...	33025.1
20	30	152160	152160	133491	4875.1	152160
30	40	18076.6	18076.6	18076.6	...	18076.6
40	50	79635.6	79635.6	72554.2	139.044	79635.6
50	60	97969.7	97969.7	85094.5	...	97969.7
60	70	84231.5	84231.5	75482.8	...	84231.5
70	80	80664.5	80664.5	67789.4	139.044	80664.5
80	90	93161.6	93161.6	80286.5	...	93161.6
90	100	77936.1	77936.1	66310.4	...	77936.1
100	110	73568.4	73568.4	61653.4	...	73568.4
110	120	70590.3	70590.3	61232.8	3138.99	70590.3
120	130	50607.7	50607.7	59039.2	...	50607.7
130	140	68270.9	68270.9	68270.9	...	68270.9

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg	Cumulative Oper. Wgt. kg	Cumulative Hydro. Wgt. kg
10	20	...	...	...
20	30	-33025.3	-33025.3	-40034.1
30	40	979898	979898	715791
40	50	776636	776636	697714
50	60	697001	697001	625160
60	70	599031	599031	540065
70	80	514800	514800	464582
80	90	434135	434135	396793
90	100	340973	340973	316507
100	110	263037	263037	250196
110	120	189469	189469	188543
120	130	118879	118879	127310
130	140	68270.9	68270.9	68270.9

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. kgf-m.	Cumulative Oper. Mom. kgf-m.	Cumulative Hydro. Mom. kgf-m.
10	20	...	...	...
20	30	-4875.1	-4875.1	-4875.1
30	40	3417.08	3417.08	3417.08
40	50	3417.08	3417.08	3417.08
50	60	3278.04	3278.04	3278.04
60	70	3278.04	3278.04	3278.04
70	80	3278.04	3278.04	3278.04
80	90	3138.99	3138.99	3138.99
90	100	3138.99	3138.99	3138.99
100	110	3138.99	3138.99	3138.99
110	120	3138.99	3138.99	3138.99
120	130	...	...	...
130	140	...	...	...





There does not appear to be any valid core flange data to process. Please review the nozzle summary for MAWP information on Jacket side nozzles if there are any.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 2.05011 Hz.

The Natural Frequency for the Vessel (Ope...) is 2.05011 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	0
Base Elevation	700 cm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 cm
Distance Upwind of Crest Lh	0 cm
Distance from Crest to the Vessel x	0 cm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 cm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.137 \cdot 0.917) / (1 + 1.7 \cdot 3.4 \cdot 0.137)))$
 $= \min(0.85, 0.891)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	2.050 Hz
Natural Frequency of Vessel (Empty)	2.050 Hz
Natural Frequency of Vessel (Test)	2.227 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.811
Structure Height to Diameter ratio	7.499
Height to top of Structure	2852.998 cm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 58978.801$ cm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 91.462 + 700.0 = 791.462$ cm
 $= 25.967$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (25.967 ft.) > 15 ft.
 $= 2.41(z / Z_g)^{2/\alpha}$
 $= 2.41(25.967 / 1935.0)^{2/11.5}$
 $= 1.139$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.139 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 36.5 \text{ psf } [178.054 \text{ kgf/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 36.468 \cdot 0.85 \cdot 0.811 \cdot 51.231$$

$$= 1288.1 \text{ lbs. } [584.3 \text{ kgf}]$$

Element	Hgt (z) cm	K1	K2	K3	Kz	Kzt	qz kgf/m ²
Bottom head	791.5	0.000	0.000	0.000	1.139	1.000	178.054
Shell-1	1067.1	0.000	0.000	0.000	1.199	1.000	187.552
Skirt	1131.7	0.000	0.000	0.000	1.212	1.000	189.480
Shell-2	1367.1	0.000	0.000	0.000	1.252	1.000	195.810
Shell-3	1599.6	0.000	0.000	0.000	1.287	1.000	201.232
Shell-4	1899.6	0.000	0.000	0.000	1.326	1.000	207.338
Shell-5	2199.6	0.000	0.000	0.000	1.360	1.000	212.694
Shell-6	2499.6	0.000	0.000	0.000	1.391	1.000	217.476
Shell-7	2799.6	0.000	0.000	0.000	1.418	1.000	221.806
Shell-8	3099.6	0.000	0.000	0.000	1.444	1.000	225.767
Shell-9	3389.7	0.000	0.000	0.000	1.466	1.000	229.308
Shell flange	3593.8	0.000	0.000	0.000	1.481	1.000	231.652
Top cover flang	3680.1	0.000	0.000	0.000	1.488	1.000	232.609

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, cm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, cm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure cm
- Lc - Total length of conical section(s) of vessel, cm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, kgf
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. kgf
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.00032 (High); 0.00032 < 0.0004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/2852.998 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[12.48/93.6^2] \cdot 10^4 = 14.246$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.0004$ no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 975910 / (2853.0 \cdot 348.728^2)$$



= 0.28128E-02

Since Vp is > 0.0004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Bottom assemb	0.00	560.00	178.05	0.00
SF weight	0.00	700.00	178.05	0.00
Sup ring wt	0.00	700.00	178.05	0.00
Skirt wt	0.00	1235.00	192.26	0.00
Cleat wt	0.00	2400.00	215.89	0.00
10MT MISC wt	0.00	2450.00	216.69	0.00
Trunnion-1	0.00	3280.00	227.97	0.00
Trunnion-2	0.00	3280.00	227.97	0.00
A4 nozzle wt	0.00	3370.00	229.07	0.00

The Natural Frequency for the Vessel (Ope...) is 2.05011 Hz.

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	791.462	381.36	47594.8	178.054	642.679
20	30	1067.1	399.092	173605	187.552	2469.26
30	40	1131.71	430.288	129086	189.48	1854.92
40	50	1367.1	415.92	68626.8	195.81	1019.08
50	60	1599.6	415.92	124776	201.232	1904.19
60	70	1899.6	415.92	124776	207.338	1961.97
70	80	2199.6	415.92	124776	212.694	2012.65
80	90	2499.6	415.92	124776	217.476	2057.9
90	100	2799.6	415.92	124776	221.806	2098.87
100	110	3099.6	415.92	124776	225.767	2136.36
110	120	3389.69	415.92	116532	229.308	2026.5
120	130	3593.85	348	44370	231.652	779.484
130	140	3680.1	480	21600	232.609	381.033

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.500
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(93.60230.75)$
 = 0.602 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/2.05)
 = 0.488

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.602$, 0.488) per 12.8.2
 = 0.488

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.217 / (1.5 / 1.5)$
 = 0.217

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.488(1.5 / 1.5))$
 = 0.190

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.2173 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.19 \cdot 979898$
 = 186158.328 kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.217 \cdot 979898$



= 42592.9 kgf

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 2.05011 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height cm	Earthquake Weight kgf	Element Ope Load kgf	Element Emp Load kgf	Element Vertical Load kgf
10	20	-0.001524	33025.1	0.0070007	0.0070007	-1435.49
20	30	217.5	152160	4603.35	4603.35	-8049.39
30	40	585	18076.6	1470.92	1470.92	42592.9
40	50	517.5	79635.6	5732.33	5732.33	33757.8
50	60	750	97969.7	10220.4	10220.4	30296.3
60	70	1050	84231.5	12302.1	12302.1	26037.9
70	80	1350	80664.5	15147.1	15147.1	22376.6
80	90	1650	93161.6	21381.3	21381.3	18870.4
90	100	1950	77936.1	21139.2	21139.2	14821
100	110	2250	73568.4	23024.4	23024.4	11433.4
110	120	2540.09	70590.3	24940.7	24940.7	8235.58
120	130	2744.25	50607.7	19317.6	19317.6	5167.26
130	140	2830.5	68270.9	26879	26879	2967.51

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support cm	Cumulative Wind Shear kgf	Earthquake Shear kgf	Wind Bending kgf-m.	Earthquake Bending kgf-m.
10	20	226.462	...	...	...	...
20	30	82.5	642.679	0.0070007	587.81	0.006403
30	40	150	21344.9	186158	255183	3410372
40	50	382.5	16378	180084	202685	2864115
50	60	615	15359	174352	176502	2571705
60	70	915	13454.8	164131	133281	2063980
70	80	1215	11492.8	151829	95859.8	1590038
80	90	1515	9480.15	136682	64400.3	1157271
90	100	1815	7422.25	115301	39046.7	779296
100	110	2115	5323.38	94161.6	19928.2	465102
110	120	2405.09	3187.02	71137.3	7162.63	217153
120	130	2608.93	1160.52	46196.6	1068.47	52633.4
130	140	2695.18	381.033	26879	85.7326	6047.78

Notes regarding vessels supported by Intermediate Skirts:

This model contains an intermediate skirt. The moment shown in the table above for the skirt, is the moment at the base of the skirt.

It is computed by summing the moments framing into the top of the skirt and then by multiplying the total shear times the skirt length.

The force on the skirt, times its moment arm is added to the result.

The moment at the top of the skirt is also computed. The stress at the skirt top is also computed and the worst of the stresses is displayed in the stress report.

The cumulative shear force on the skirt is generally the sum of the shear forces in the wind/seismic report including the shear force on the skirt and possibly other applied forces.

Bending Moments at top of Intermediate Skirt (Wind condition):

= [-Moment at Skirt top (below) + Moment at Skirt top (above)]
 = [-8754.1 + 202684.9]
 = 193930.8 kgf-m.

Moment at Skirt Top, operating case : 193930.8 kgf-m.

Bending Moments at top of Intermediate Skirt (Earthquake condition):

= [-Moment at Skirt top (below) + Moment at Skirt top (above)]
 = [-10012.3 + 2864116]
 = 2854103.0 kgf-m.

Moment at Skirt Top, operating case : 2854103.0 kgf-m.

Moment at Skirt Top, empty case : 2854103.0 kgf-m.

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear kgf	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	...	...	...
20	30	...	...	...	...
30	40	21344.9	0.0091511	0.038883	0.27393E-04
40	50	16378	0.017968	0.00027429	0.19585E-04
50	60	15359	-0.024356	-0.040315	0.80141E-05
60	70	13454.8	-0.048776	-0.050833	0.53437E-06
70	80	11492.8	-0.047496	-0.03969	0.65011E-05
80	90	9480.15	-0.028259	-0.013956	0.10338E-04
90	100	7422.25	0.0025482	0.020673	0.12509E-04
100	110	5323.38	0.039925	0.059901	0.13484E-04
110	120	3187.02	0.078932	0.098136	0.13738E-04
120	130	1160.52	0.10689	0.11565	0.13740E-04
130	140	381.033	0.11874	0.12184	0.13740E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	140.351	877.191
20	30	146.877	917.978
30	40	158.357	989.733
40	50	153.07	956.685
50	60	153.07	956.685
60	70	153.07	956.685
70	80	153.07	956.685
80	90	153.07	956.685
90	100	153.07	956.685
100	110	153.07	956.685
110	120	153.07	956.685
120	130	128.073	800.457
130	140	176.653	1104.08

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 142.650 Actual deflection : 0.122 mm

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	9013.39	8872.33	675428224	665483968
20	30	17724.5	17450.9	1294114816	1276465152
30	40	6283.02	5974.28	513302528	488945728
40	50	17724.5	17587.8	1294114816	1285303552
50	60	17724.5	17587.8	1294115456	1285304192
60	70	17724.5	17587.8	1294115456	1285304192
70	80	17724.5	17587.8	1294115456	1285304192
80	90	17724.5	17587.8	1294115456	1285304192
90	100	17724.5	17587.8	1294115456	1285304192
100	110	17724.5	17587.8	1294115456	1285304192
110	120	17724.5	17587.8	1294115456	1285304192
120	130	17724.5	17587.8	1294115456	1285304192
130	140	17724.5	17587.8	1294115456	1285304192

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Longitudinal Allowable Stresses:

From	To	Tensile kgf/cm ²	Hydrotest Tensile kgf/cm ²	Compressive kgf/cm ²	Hydrotest Compressive kgf/cm ²
10	20	1680.42	3002.91	-948.468	-1243.06
20	30	1680.42	3002.91	-1038.66	-1283.57
30	40	1680.42	3002.91	-872.609	-1211.79
40	50	1680.42	3002.91	-1039.8	-1283.57
50	60	1680.42	3002.91	-1039.8	-1283.57
60	70	1680.42	3002.91	-1039.8	-1283.57
70	80	1680.42	3002.91	-1039.8	-1283.57
80	90	1680.42	3002.91	-1039.8	-1283.57
90	100	1680.42	3002.91	-1039.8	-1283.57
100	110	1680.42	3002.91	-1039.8	-1283.57
110	120	1680.42	3002.91	-1039.8	-1283.57
120	130	1680.42	3002.91	-1039.8	-1283.57
130	140	1680.42	3002.91	-1039.8	-1283.57

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have been computed in the new and cold condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure kgf/cm ²	Longitudinal Stress External Pressure kgf/cm ²	Longitudinal Stress Hydrotest Pressure kgf/cm ²
10	20	1546.48	...	2422.77
20	30	740.22	...	1157.44
30	40	...	...	...
40	50	732.943	...	1157.44
50	60	732.943	...	1157.44
60	70	732.943	...	1157.44
70	80	732.943	...	1157.44
80	90	732.943	...	1157.44
90	100	732.943	...	1157.44
100	110	732.943	...	1157.44
110	120	732.943	...	1157.44
120	130	...	...	...
130	140	...	...	...

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty kgf/cm ²	Wght. Str. Operating kgf/cm ²	Wght. Str. Hydrotest kgf/cm ²	Wght. Str. Emp. Mom. kgf/cm ²	Wght. Str. Opr. Mom. kgf/cm ²
10	20	...	...	...	...	...
20	30	1.89247	1.89247	...	-0.38192	-0.38192
30	40	-129.997	-129.997	...	0.69887	0.69887
40	50	-44.1578	-44.1578	...	0.26586	0.26586
50	60	-39.6298	-39.6298	...	0.25504	0.25504
60	70	-34.0595	-34.0595	...	0.25504	0.25504
70	80	-29.2703	-29.2703	...	0.25504	0.25504
80	90	-24.6839	-24.6839	...	0.24422	0.24422
90	100	-19.387	-19.387	...	0.24422	0.24422
100	110	-14.9557	-14.9557	...	0.24422	0.24422
110	120	-10.7728	-10.7728	...	0.24422	0.24422
120	130	-6.75916	-6.75916	...	...	...
130	140	-3.88172	-3.88172	...	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. kgf/cm ²	Bend. Str. Oper. Wind kgf/cm ²	Bend. Str. Oper. Equ. kgf/cm ²	Bend. Str. Hyd. Wind kgf/cm ²	Bend. Str. Hyd. Equ. kgf/cm ²
10	20	...	...	...	...	...
20	30	...	0.04605	0.50162E-06	...	...
30	40	...	39.6631	583.726	...	...
40	50	...	15.7694	222.836	...	...
50	60	...	13.7323	200.085	...	...
60	70	...	10.3696	160.583	...	...
70	80	...	7.45814	123.709	...	...
80	90	...	5.01051	90.0387	...	...
90	100	...	3.03794	60.6313	...	...
100	110	...	1.55047	36.1862	...	...
110	120	...	0.55727	16.8951	...	...
120	130	...	0.08313	4.09502	...	...
130	140	...	0.0066702	0.47053	...	...



Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case kgf/cm ²	Vortex Shedding Empty Case kgf/cm ²	Vortex Shedding Test Case kgf/cm ²	Earthquake Empty Case kgf/cm ²
10	20	...	...	...	...
20	30	...	...	...	0.50162E-06
30	40	...	...	...	...
40	50	...	...	...	222.836
50	60	...	...	...	200.085
60	70	...	...	...	160.583
70	80	...	...	...	123.709
80	90	...	...	...	90.0387
90	100	...	...	...	60.6313
100	110	...	...	...	36.1862
110	120	...	...	...	16.8951
120	130	...	...	...	4.09502
130	140	...	...	...	0.47053

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind kgf/cm ²	Longitudinal Stress Y Forces Seismic kgf/cm ²
10	20	...	-0.16179
20	30	...	-0.46126
30	40	...	7.12939
40	50	...	1.91939
50	60	...	1.72258
60	70	...	1.48045
70	80	...	1.27228
80	90	...	1.07293
90	100	...	0.84269
100	110	...	0.65007
110	120	...	0.46826
120	130	...	0.2938
130	140	...	0.16873

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded kgf/cm ²	Earthquake For/Mom Corroded kgf/cm ²	Wind For/Mom No Corrosion kgf/cm ²	Earthquake For/Mom No Corrosion kgf/cm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	583.726	...	556.028
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...
120	130	...	...	...	...
130	140	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress kgf/cm ²
------	----	--



10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Un-Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1680.42	0.00	948.47	0.0000	-0.0000	...
20	2.23	1680.42		2011.77	0.0013		0.0011
30	170.36	1680.42	-170.36	1769.65	0.1014	0.0963	0.0840
40	60.19	1680.42	-60.19	2012.63	0.0358	0.0299	0.0260
50	53.62	1680.42	-53.62	2012.63	0.0319	0.0266	0.0233
60	44.68	1680.42	-44.68	2012.63	0.0266	0.0222	0.0200
70	36.98	1680.42	-36.98	2012.63	0.0220	0.0184	0.0172
80	29.94	1680.42	-29.94	2012.63	0.0178	0.0149	0.0145
90	22.67	1680.42	-22.67	2012.63	0.0135	0.0113	0.0114
100	16.75	1680.42	-16.75	2012.63	0.0100	0.0083	0.0088
110	11.57	1680.42	-11.57	2012.63	0.0069	0.0058	0.0063

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.16	1680.42	-0.16	1116.89	0.0001	0.0001	0.0101



20	1.81	1680.42		2011.77	0.0011	...	
30	707.29	1680.42	-707.29	1769.65	0.4209	0.3997	0.3302
40	265.34	1680.42	-265.34	2012.63	0.1579	0.1318	0.1109
50	238.25	1680.42	-238.25	2012.63	0.1418	0.1184	0.0995
60	193.42	1680.42	-193.42	2012.63	0.1151	0.0961	0.0799
70	151.96	1680.42	-151.96	2012.63	0.0904	0.0755	0.0616
80	113.89	1680.42	-113.89	2012.63	0.0678	0.0566	0.0449
90	79.42	1680.42	-79.42	2012.63	0.0473	0.0395	0.0302
100	50.74	1680.42	-50.74	2012.63	0.0302	0.0252	0.0181
110	27.44	1680.42	-27.44	2012.63	0.0163	0.0136	0.0118

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1680.42	0.00	1512.08	0.0000	-0.0000	0.0180
20	2.23	1680.42		2011.77	0.0013		0.0011
30	170.36	1680.42	-170.36	1769.65	0.1014	0.0963	0.0840
40	60.19	1680.42	-60.19	2012.63	0.0358	0.0299	0.0260
50	53.62	1680.42	-53.62	2012.63	0.0319	0.0266	0.0233
60	44.68	1680.42	-44.68	2012.63	0.0266	0.0222	0.0200
70	36.98	1680.42	-36.98	2012.63	0.0220	0.0184	0.0172
80	29.94	1680.42	-29.94	2012.63	0.0178	0.0149	0.0145
90	22.67	1680.42	-22.67	2012.63	0.0135	0.0113	0.0114
100	16.75	1680.42	-16.75	2012.63	0.0100	0.0083	0.0088
110	11.57	1680.42	-11.57	2012.63	0.0069	0.0058	0.0063

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.16	1680.42	-0.16	1116.89	0.0001	0.0001	0.0101
20	1.81	1680.42		2011.77	0.0011		...
30	1291.02	1680.42	-1291.02	1769.65	0.7683	0.7295	0.6601
40	265.34	1680.42	-265.34	2012.63	0.1579	0.1318	0.1109
50	238.25	1680.42	-238.25	2012.63	0.1418	0.1184	0.0995
60	193.42	1680.42	-193.42	2012.63	0.1151	0.0961	0.0799
70	151.96	1680.42	-151.96	2012.63	0.0904	0.0755	0.0616
80	113.89	1680.42	-113.89	2012.63	0.0678	0.0566	0.0449
90	79.42	1680.42	-79.42	2012.63	0.0473	0.0395	0.0302
100	50.74	1680.42	-50.74	2012.63	0.0302	0.0252	0.0181
110	27.44	1680.42	-27.44	2012.63	0.0163	0.0136	0.0118

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	3002.91	0.00	1243.06	0.0000	-0.0000	0.0180
20	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
30	0.00	3002.91	0.00	1211.79	0.0000	-0.0000	0.0017
40	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
50	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
60	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
70	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
80	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	0.0012
90	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	0.0013
100	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	0.0015
110	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	0.0017

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	3002.91	0.00	1243.06	0.0000	-0.0000	...
20	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
30	0.00	3002.91	0.00	1211.79	0.0000	-0.0000	...
40	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
50	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
60	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
70	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...



80	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
90	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
100	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...
110	0.00	3002.91	0.00	1283.57	0.0000	-0.0000	...

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.48	1680.42		948.47	0.9203		...
20	1362.23	1680.42		2011.77	0.8106		0.0011
30	170.36	1680.42	-170.36	1769.65	0.1014	0.0963	0.0840
40	1353.03	1680.42		2012.63	0.8052		0.0260
50	1352.53	1680.42		2012.63	0.8049		0.0233
60	1351.90	1680.42		2012.63	0.8045		0.0200
70	1351.40	1680.42		2012.63	0.8042		0.0172
80	1350.99	1680.42		2012.63	0.8040		0.0145
90	1350.60	1680.42		2012.63	0.8037		0.0114
100	1350.32	1680.42		2012.63	0.8036		0.0088
110	1350.09	1680.42		2012.63	0.8034		0.0063

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.40	1680.42		948.47	0.9202		...
20	1362.25	1680.42		2011.77	0.8107		...
30	1291.02	1680.42	-1291.02	1769.65	0.7683	0.7295	0.6601
40	1384.29	1680.42		2012.63	0.8238		0.1109
50	1378.45	1680.42		2012.63	0.8203		0.0995
60	1369.92	1680.42		2012.63	0.8152		0.0799
70	1363.29	1680.42		2012.63	0.8113		0.0616
80	1358.29	1680.42		2012.63	0.8083		0.0449
90	1354.67	1680.42		2012.63	0.8061		0.0302
100	1352.32	1680.42		2012.63	0.8047		0.0181
110	1350.85	1680.42		2012.63	0.8039		0.0118

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1680.42	0.00	948.47	0.0000	-0.0000	...
20	2.23	1680.42		2021.19	0.0013		0.0011
30	170.36	1680.42	-170.36	1801.36	0.1014	0.0946	0.0819
40	60.19	1680.42	-60.19	2021.98	0.0358	0.0298	0.0255
50	53.62	1680.42	-53.62	2021.98	0.0319	0.0265	0.0229
60	44.68	1680.42	-44.68	2021.98	0.0266	0.0221	0.0197
70	36.98	1680.42	-36.98	2021.98	0.0220	0.0183	0.0169
80	29.94	1680.42	-29.94	2021.98	0.0178	0.0148	0.0143
90	22.67	1680.42	-22.67	2021.98	0.0135	0.0112	0.0112
100	16.75	1680.42	-16.75	2021.98	0.0100	0.0083	0.0087
110	11.57	1680.42	-11.57	2021.98	0.0069	0.0057	0.0062

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.16	1680.42	-0.16	1123.27	0.0001	0.0001	0.0101
20	1.81	1680.42		2021.19	0.0011		...
30	1291.02	1680.42	-1291.02	1801.36	0.7683	0.7167	0.6485
40	265.34	1680.42	-265.34	2021.98	0.1579	0.1312	0.1103
50	238.25	1680.42	-238.25	2021.98	0.1418	0.1178	0.0991
60	193.42	1680.42	-193.42	2021.98	0.1151	0.0957	0.0795
70	151.96	1680.42	-151.96	2021.98	0.0904	0.0752	0.0613
80	113.89	1680.42	-113.89	2021.98	0.0678	0.0563	0.0447
90	79.42	1680.42	-79.42	2021.98	0.0473	0.0393	0.0301
100	50.74	1680.42	-50.74	2021.98	0.0302	0.0251	0.0180
110	27.44	1680.42	-27.44	2021.98	0.0163	0.0136	0.0118

Analysis of Load Case 11 : HP+HW+HI



From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	2503.68	3002.91		1243.06	0.8338		0.0180
20	2270.17	3002.91		1283.57	0.7560		...
30	0.00	3002.91	0.00	1211.79	0.0000	-0.0000	0.0017
40	2270.17	3002.91		1283.57	0.7560		...
50	2270.17	3002.91		1283.57	0.7560		...
60	2270.17	3002.91		1283.57	0.7560		...
70	2270.17	3002.91		1283.57	0.7560		...
80	2270.17	3002.91		1283.57	0.7560		0.0012
90	2270.17	3002.91		1283.57	0.7560		0.0013
100	2270.17	3002.91		1283.57	0.7560		0.0015
110	2270.17	3002.91		1283.57	0.7560		0.0017

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	2503.68	3002.91		1243.06	0.8338		...
20	2270.17	3002.91		1283.57	0.7560		...
30	0.00	3002.91	0.00	1211.79	0.0000	-0.0000	...
40	2270.17	3002.91		1283.57	0.7560		...
50	2270.17	3002.91		1283.57	0.7560		...
60	2270.17	3002.91		1283.57	0.7560		...
70	2270.17	3002.91		1283.57	0.7560		...
80	2270.17	3002.91		1283.57	0.7560		...
90	2270.17	3002.91		1283.57	0.7560		...
100	2270.17	3002.91		1283.57	0.7560		...
110	2270.17	3002.91		1283.57	0.7560		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.48	1680.42		948.47	0.9203		...
20	1362.23	1680.42		2011.77	0.8106		0.0011
30	130.70	1680.42	-130.70	1769.65	0.0778	0.0739	0.0840
40	1351.88	1680.42		2012.63	0.8045		0.0260
50	1351.59	1680.42		2012.63	0.8043		0.0233
60	1351.24	1680.42		2012.63	0.8041		0.0200
70	1350.97	1680.42		2012.63	0.8039		0.0172
80	1350.72	1680.42		2012.63	0.8038		0.0145
90	1350.45	1680.42		2012.63	0.8036		0.0114
100	1350.24	1680.42		2012.63	0.8035		0.0088
110	1350.06	1680.42		2012.63	0.8034		0.0063

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.48	1680.42		948.47	0.9203		...
20	1362.22	1680.42		2011.77	0.8106		0.0011
30	123.61	1680.42	-123.61	1769.65	0.0736	0.0698	0.0799
40	1351.84	1680.42		2012.63	0.8045		0.0258
50	1351.55	1680.42		2012.63	0.8043		0.0231
60	1351.21	1680.42		2012.63	0.8041		0.0199
70	1350.94	1680.42		2012.63	0.8039		0.0171
80	1350.70	1680.42		2012.63	0.8038		0.0144
90	1350.43	1680.42		2012.63	0.8036		0.0113
100	1350.23	1680.42		2012.63	0.8035		0.0087
110	1350.05	1680.42		2012.63	0.8034		0.0063

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.48	1680.42		948.47	0.9203		...
20	1362.23	1680.42		2011.77	0.8106		0.0011
30	130.70	1680.42	-130.70	1769.65	0.0778	0.0739	0.0840



40	1351.88	1680.42	2012.63	0.8045	0.0260
50	1351.59	1680.42	2012.63	0.8043	0.0233
60	1351.24	1680.42	2012.63	0.8041	0.0200
70	1350.97	1680.42	2012.63	0.8039	0.0172
80	1350.72	1680.42	2012.63	0.8038	0.0145
90	1350.45	1680.42	2012.63	0.8036	0.0114
100	1350.24	1680.42	2012.63	0.8035	0.0088
110	1350.06	1680.42	2012.63	0.8034	0.0063

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.48	1680.42		948.47	0.9203		...
20	1362.23	1680.42		2011.77	0.8106		0.0011
30	130.70	1680.42	-130.70	1769.65	0.0778	0.0739	0.0840
40	1351.88	1680.42		2012.63	0.8045		0.0260
50	1351.59	1680.42		2012.63	0.8043		0.0233
60	1351.24	1680.42		2012.63	0.8041		0.0200
70	1350.97	1680.42		2012.63	0.8039		0.0172
80	1350.72	1680.42		2012.63	0.8038		0.0145
90	1350.45	1680.42		2012.63	0.8036		0.0114
100	1350.24	1680.42		2012.63	0.8035		0.0088
110	1350.06	1680.42		2012.63	0.8034		0.0063

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.00	1680.42	0.00	948.47	0.0000	-0.0000	...
20	2.27	1680.42		2011.77	0.0014		0.0011
30	130.70	1680.42	-130.70	1769.65	0.0778	0.0739	0.0840
40	44.42	1680.42	-44.42	2012.63	0.0264	0.0221	0.0260
50	39.88	1680.42	-39.88	2012.63	0.0237	0.0198	0.0233
60	34.31	1680.42	-34.31	2012.63	0.0204	0.0170	0.0200
70	29.53	1680.42	-29.53	2012.63	0.0176	0.0147	0.0172
80	24.93	1680.42	-24.93	2012.63	0.0148	0.0124	0.0145
90	19.63	1680.42	-19.63	2012.63	0.0117	0.0098	0.0114
100	15.20	1680.42	-15.20	2012.63	0.0090	0.0076	0.0088
110	11.02	1680.42	-11.02	2012.63	0.0066	0.0055	0.0063

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.40	1680.42		1116.89	0.9202		0.0096
20	1362.25	1680.42		2011.77	0.8107		...
30	707.29	1680.42	-707.29	1769.65	0.4209	0.3997	0.3299
40	1351.76	1680.42		2012.63	0.8044		0.0249
50	1351.48	1680.42		2012.63	0.8042		0.0223
60	1351.16	1680.42		2012.63	0.8041		0.0192
70	1350.90	1680.42		2012.63	0.8039		0.0165
80	1350.66	1680.42		2012.63	0.8038		0.0139
90	1350.41	1680.42		2012.63	0.8036		0.0109
100	1350.22	1680.42		2012.63	0.8035		0.0084
110	1350.04	1680.42		2012.63	0.8034		0.0061

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	0.16	1680.42	-0.16	948.47	0.0001	0.0002	...
20	1.81	1680.42		2021.19	0.0011		...
30	707.29	1680.42	-707.29	1801.36	0.4209	0.3926	0.3240
40	42.50	1680.42	-42.50	2021.98	0.0253	0.0210	0.0244
50	38.16	1680.42	-38.16	2021.98	0.0227	0.0189	0.0219
60	32.83	1680.42	-32.83	2021.98	0.0195	0.0162	0.0188
70	28.25	1680.42	-28.25	2021.98	0.0168	0.0140	0.0162
80	23.86	1680.42	-23.86	2021.98	0.0142	0.0118	0.0137
90	18.79	1680.42	-18.79	2021.98	0.0112	0.0093	0.0107



100	14.55	1680.42	-14.55	2021.98	0.0087	0.0072	0.0083
110	10.55	1680.42	-10.55	2021.98	0.0063	0.0052	0.0060

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
10	1546.48	1680.42		1116.89	0.9203		0.0092
20	1362.23	1680.42		2011.77	0.8106		0.0011
30	130.70	1680.42	-130.70	1769.65	0.0778	0.0739	0.0840
40	1351.88	1680.42		2012.63	0.8045		0.0260
50	1351.59	1680.42		2012.63	0.8043		0.0233
60	1351.24	1680.42		2012.63	0.8041		0.0200
70	1350.97	1680.42		2012.63	0.8039		0.0172
80	1350.72	1680.42		2012.63	0.8038		0.0145
90	1350.45	1680.42		2012.63	0.8036		0.0114
100	1350.24	1680.42		2012.63	0.8035		0.0088
110	1350.06	1680.42		2012.63	0.8034		0.0063

Absolute Maximum of the all of the Stress Ratio's 0.9203

Governing Element: Bottom head

Governing Load Case 7 : IP+OW+WI+FW+BW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom head	1546.5	...	...	...	1546.5	+ Tens
Bottom head	1546.5	...	...	...	...	- Comp
Shell-1	740.2	1.9	-0.4	...	741.8	+ Tens
Shell-1	740.2	1.9	0.4	...	...	- Comp
Skirt	...	-130.0	0.7	39.7	...	+ Tens
Skirt	...	-130.0	-0.7	-39.7	-170.4	- Comp
Shell-2	732.9	-44.2	0.3	15.8	704.8	+ Tens
Shell-2	732.9	-44.2	-0.3	-15.8	...	- Comp
Shell-3	732.9	-39.6	0.3	13.7	707.3	+ Tens
Shell-3	732.9	-39.6	-0.3	-13.7	...	- Comp
Shell-4	732.9	-34.1	0.3	10.4	709.5	+ Tens
Shell-4	732.9	-34.1	-0.3	-10.4	...	- Comp
Shell-5	732.9	-29.3	0.3	7.5	711.4	+ Tens
Shell-5	732.9	-29.3	-0.3	-7.5	...	- Comp
Shell-6	732.9	-24.7	0.2	5.0	713.5	+ Tens
Shell-6	732.9	-24.7	-0.2	-5.0	...	- Comp
Shell-7	732.9	-19.4	0.2	3.0	716.8	+ Tens
Shell-7	732.9	-19.4	-0.2	-3.0	...	- Comp
Shell-8	732.9	-15.0	0.2	1.6	719.8	+ Tens
Shell-8	732.9	-15.0	-0.2	-1.6	...	- Comp
Shell-9	732.9	-10.8	0.2	0.6	723.0	+ Tens
Shell-9	732.9	-10.8	-0.2	-0.6	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length cm
- Fxa - Allowable comp mem stress due to compression with lamc <= 0.15, kgf/cm²
- Fba - Allowable all comp mem stress due to bending moment, kgf/cm²
- Fva - Allowable shear stress when only shear is present, kgf/cm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, kgf/cm²



Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , kgf/cm²

Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , kgf/cm²

Faha - Allowable axial comp w/hoop compression and lamc > 0.15, kgf/cm²

Fbha - Allowable axial comp from bending w/hoop compression, kgf/cm²

Fhba - Allowable hoop comp w/long compression due to moment, kgf/cm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

[illegible]

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

[illegible]

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

[illegible]

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

[illegible]



Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1512	70307		70307	70307	70307
20	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
30	0.620	27300	1815	1815	943		70307	1815	70307	70307	70307
40	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
50	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
60	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
70	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
80	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
90	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
100	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
110	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
30	0.620	27300	1815	1815	943		70307	1815	70307	70307	70307
40	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
50	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
60	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
70	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
80	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
90	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
100	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
110	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.700	27300	2021	2021	1349	988	70307	2021	70307	70307	70307
30	0.614	27300	1801	1801	926		70307	1801	70307	70307	70307
40	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307



50	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
60	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
70	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
80	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
90	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
100	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
110	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1123	70307		70307	70307	70307
20	0.700	27300	2021	2021	1349	988	70307	2021	70307	70307	70307
30	0.614	27300	1801	1801	926		70307	1801	70307	70307	70307
40	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
50	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
60	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
70	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
80	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
90	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
100	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
110	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1512	70307		70307	70307	70307
20	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
30	0.620	27300	1815	1815	943		70307	1815	70307	70307	70307
40	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
50	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
60	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
70	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
80	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
90	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
100	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
110	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
30	0.620	27300	1815	1815	943		70307	1815	70307	70307	70307
40	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
50	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
60	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
70	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
80	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
90	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
100	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
110	0.704	27300	2021	2021	1355	996	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307



110 0.732 27300 2012 2012 1349 983 70307 2012 70307 70307 70307
 120
 130

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
------	------	---	-----	-----	-----	-----	------	------	------	------	------



10						1116	70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10							70307		70307	70307	70307
20	0.700	27300	2021	2021	1349	988	70307	2021	70307	70307	70307
30	0.614	27300	1801	1801	926		70307	1801	70307	70307	70307
40	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
50	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
60	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
70	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
80	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
90	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
100	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
110	0.700	27300	2021	2021	1352	992	70307	2021	70307	70307	70307
120											
130											

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
10						1116	70307		70307	70307	70307
20	0.732	27300	2011	2011	1345	979	70307	2011	70307	70307	70307
30	0.638	27300	1769	1769	916		70307	1769	70307	70307	70307
40	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
50	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
60	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
70	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
80	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
90	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
100	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
110	0.732	27300	2012	2012	1349	983	70307	2012	70307	70307	70307
120											
130											

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.



Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt

Mean skirt diameter at the critical height is computed later.
 This is a Conical type Support Skirt , Wider at the Bottom

Half Apex Angle of the skirt	a	3.814 deg
Total vessel (empty) weight	Fg	961821.562 kgf
Weight of vessel contents (liquid)	Ff	0.000 kgf
Vessel Gross Weight	Fg + Ff	961821.562 kgf
Skirt Mean Diameter Top End	DzTop	3328.874 mm
Skirt Mean Diameter Bottom End	DzBot	3728.874 mm
Skirt Analysis Thickness (Corroded)	ez	57.000 mm
Skirt Height	H	3000.000 mm
Additional Axial Force at section 1-1	F1	0.000 kgf
Moment at section 1-1	M1	2864115.500 kgf-m.
Applied Moment - Bottom of Skirt	Mb	3410372.500 kgf-m.
Weight at Bottom of Skirt	W	979898.312 kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:



For a Type B Skirt, the next element must be a dished head and this is not the case. Please refer to Figure 16.12-2 of EN 13445-3 for more information.

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Shop/Field Installation Options :

Packing is installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	1165.944 cm
Center of Gravity of the Insulation	1475.896 cm
Center of Gravity of the Nozzles	840.230 cm
Center of Gravity of the Added Weights (Operating)	580.841 cm
Center of Gravity of the Added Weights (Empty)	580.841 cm
Center of Gravity of Bare Shell New and Cold	1613.175 cm
Center of Gravity of Bare Shell Corroded	1616.955 cm
Vessel CG in the Operating Condition	1361.062 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	1360.107 cm
Vessel CG in the Test Condition	1358.448 cm

Rigging Analysis Results:

Because this model is supported by an intermediate skirt, the rigging analysis cannot be performed.

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**Skirt Data :**

Skirt Outside Diameter at Base	SOD, D6	3785.7336	mm
Skirt Thickness	STHK	60.0000	mm
Skirt Internal Corrosion Allowance	SCA	3.0000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-387 11	[Impact Tested]

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	44.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-387 11	
Basing Ope. All. Stress	BASOPE, f4	2110.70	kgf/cm ²
Basing Yield Stress		3160.95	kgf/cm ²
Inside Diameter of Basing	DI, D4	3346.0000	mm
Outside Diameter of Basing	DOU, D5	4054.0002	mm
Nominal Diameter of Bolts	BND, d(b0)	48.0000	mm
Bolt Corrosion Allowance	BCA	0.7500	mm
Root Area of a Single Bolt	Area	12.2480	cm ²
Bolt Material		SA-193 B7M	
Bolt Material Class/Thickness Range		<= 100	
Bolt Operating Allowable Stress	SA, fB	3163.82	kgf/cm ²
Number of Bolts	RN, nb	40	
Diameter of Bolt Circle	DC, DBC	3925.9998	mm
Bolt Allowable Shear Stress		2109.139	kgf/cm ²
Ultimate Comp. Strength of Concrete	FPC	356.9	kgf/cm ²
Allowable Comp. Strength of Concrete	FC	160.6	kgf/cm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	42.0000	mm
Width of Gussets at Top Plate	TWDT	134.0000	mm
Width of Gussets at Base Plate	BWDT	134.0000	mm
Gusset Plate Elastic Modulus	E	2071226.9	kgf/cm ²
Gusset Plate Yield Stress	SY	3160.9	kgf/cm ²
Height of Chair	h1	280.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	39.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	134.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	60.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	979898.2	kgf
Operating Weight of Vessel	ROW	979898.3	kgf
Earthquake Moment on Basing	EQMOM	3412247.2	kgf-m.
Wind Moment on Basing	WIMOM	257058.3	kgf-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 3412247/3926.0) - 937305) / 40$$

$$= 63481.4 \text{ kgf}$$



Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 257058/3926.0) - 979898) / 40$$

$$= -17949.9 \text{ kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3926.0) - 979898) / 40$$

$$= -24497.5 \text{ kgf}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	12.2480	cm ²
Area Available in all the Bolts	Abss * RN :	489.9200	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.157	0.235	1.072	2.758	0.468	0.772	557513.4	1494818.8	16727.8
4	0.274	0.293	1.437	2.499	0.443	0.780	581623.4	1518928.8	19257.2
6	0.303	0.298	1.518	2.435	0.437	0.781	588207.4	1525512.9	19985.5
8	0.296	0.297	1.498	2.451	0.439	0.781	586541.8	1523847.2	19799.2
10	0.298	0.298	1.503	2.447	0.438	0.781	586956.8	1524262.2	19845.5
11	0.298	0.298	1.504	2.447	0.438	0.781	587026.2	1524331.5	19853.3

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 587026 / (3.972 \cdot 3926.0 \cdot 2.447)$$

$$= 3077.265 \text{ kgf/cm}^2, \text{ Should be } \leq 3163.8 \text{ kgf/cm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 40 \cdot 12.248 / (\pi \cdot 3926.0)$$

$$= 3.972 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 132.715[(2 \cdot 0.298 \cdot 3926.0 + 354.0) / (2 \cdot 0.298 \cdot 3926.0)]$$

$$= 152.797 \text{ kgf/cm}^2, \text{ Should be } \leq 160.6 \text{ kgf/cm}^2$$

Values for table 10.3, l = 134.133, b = 154.174, l/b = 0.870014

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(2943.881, 4429.167) = 4429.167 \text{ kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 4429.167 / 2086.2)} + 1.5$$

$$= 37.191 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 134.0, b = 102.0, b/l = 0.761194

As b/l (0.761) is less than 1, inverting b/l = 1.314.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2 \cdot \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4 \cdot \pi)]$$

$$= 38750.41 / (4 \cdot \pi) [1.3(\ln((2 \cdot 134.0 \cdot \sin(\pi \cdot 70.133/134.0)) / (\pi \cdot 39.688))) + 1] - [(0.7 - 0.093) \cdot 38750.41 / (4 \cdot \pi)]$$

$$= 4269.4648 \text{ kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$



$$= \sqrt{6 \cdot \text{Abs}(4269.46)/2107.3} + 1.5$$

$$= 36.3658 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot \text{lavg}) + \text{Ca}$$

$$= 38750.41 / (1265.526 \cdot 134.0) + 1.5$$

$$= 24.351 \text{ (not less than } 9.525 + 1.5 \text{) mm}$$

Bolt spacing [m]:

$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3926.0 / 40 = 308.347 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 (P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot \text{Sktope}))^{2/3} \cdot r^{1/3} + \text{Ca}$$

$$= 1.76 (38750 \cdot 70.133 / (308.35 \cdot 241.0 \cdot 2521))^{2/3} \cdot 1892.871^{1/3} + \text{Ca}$$

$$= 27.902 + 3.0 = 30.902 \text{ mm}$$

Shear Stress in a Single Bolt [rb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 186158.3 / (12.248 \cdot 40)$$

$$= 380.0 \text{ kgf/cm}^2. \text{ Must be less than } 2109.1 \text{ kgf/cm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	37.1908	mm
Actual Basing Thickness as entered by user	44.0000	mm
Required Thickness of Chair Cap	36.3658	mm
Actual Top Ring Thickness as entered by user	39.0000	mm
Required Gusset thickness, + CA	24.3508	mm
Actual Gusset Thickness as entered by user	42.0000	mm
Required Thickness of Skirt for Local Stress	30.9024	mm
Given Thickness of Skirt	60.0000	mm
Required Gusset Height to meet local stress	35.3862	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - \text{CA})$$

$$= 1291.019 \cdot (60.0 - 3.0)$$

$$= 735.88 \text{ kgf/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 736 / [(0.4 \cdot 2521) \cdot 2 \cdot 0.707]$$

$$= 51.617 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 38750.4 / (210.0 + 2 \cdot (197.0 + 39.0))$$

$$= 56.819 \text{ kgf/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 38750.4 \cdot 70.133 / (210.0 \cdot (236.0) + 0.6667 \cdot (236.0)^2)$$

$$= 31.349 \text{ kgf/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{56.82^2 + 31.35^2}$$

$$= 64.893 \text{ kgf/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 64.89 / [(0.4 \cdot 2521) \cdot 2 \cdot 0.707]$$

$$= 4.552 \text{ mm}$$



Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
 = Bolt Load / (2 • TopWth)
 = 38750.4/(2 • 134.0)
 = 144.591 kgf/mm

Weld Load [Wh]:
 = Bolt Load • e / (2 • Hgt • TopWth)
 = 38750.4 • 70.13/(2 • 236.0 • 134.0)
 = 42.969 kgf/mm

Resultant Weld Load [Wr]:
 = $\sqrt{Wv^2 + Wh^2}$
 = $\sqrt{144.59^2 + 42.97^2}$
 = 150.841 kgf/mm

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
 = Wr / [(0.4 • Yield) • 2 • 0.707]
 = 150.84/[(0.4 • 2521) • 2 • 0.707]
 = 10.58 mm

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	51.6165	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	10.5804	mm
Required Gusset to Top Plate Double Fillet Weld Size	10.5804	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: A2 12"

From: 10

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Hemispherical Head	D	2992.00	mm
Head Finished (Minimum) Thickness	t	93.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1000.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

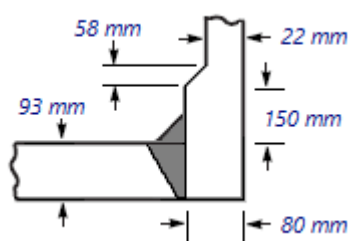
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		300.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	58.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	93.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		45.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A2 12"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	300.000	mm.
Actual Thickness Used in Calculation	22.000	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 303.0 \cdot (\exp(194.759/(1680.42 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 18.6167 + 1.5 + 0.0 = 20.1167 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1680.42 \cdot 1.0 \cdot \ln((2 \cdot 20.5 + 303.0) / 303.0) - 0.0 \\ &= 213.26 - 0.0 = 213.2601 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 9.840 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 90.869 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 9.840 \text{ mm}$$

Minimum nozzle neck thickness 21.996 mm, is greater than required thickness 9.840 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1680.4/1680.4, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness - corrosion allowance} \\ &= 80.0 - 1.5 \\ &= 78.500 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness - corrosion allowance} \\ &= 22.0 - 1.5 \\ &= 20.500 \text{ mm} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 2992.0/2 + 1.5 \\ &= 1497.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{1497.5 \cdot 91.5}, 2 \cdot 151.5) \\ &= 303.000 \text{ mm} \end{aligned}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((91.5 + 0.0)/78.5)^{0.35}, 1) \\ &= \min(1.055, 1) \\ &= 1.000 \end{aligned}$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t) \\ &= \min(91.5 + 0.0 + 1.0 \cdot \sqrt{151.5 \cdot 78.5}, 500.0 + 91.5) \\ &= 200.554 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 91.5 + ((0.0 \cdot 1.0) / 303.0) \\ &= 91.500 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :



Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 91.5 \cdot 303.0 \\ &= 277.245 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 78.5 \cdot 200.554 \\ &= 157.435 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 277.245 + 1 \cdot (157.435 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 435.179 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 78.5 / \ln[1 + 78.5/151.5] \\ &= 188.027 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{\text{eff}} / \ln[1 + t_{\text{eff}}/R_{\text{eff}}] \\ &= 91.5 / \ln[1 + 91.5/1497.5] \\ &= 1542.799 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{\text{xn}} \cdot LH \\ &= 194.759 \cdot 188.027 \cdot 200.554 \\ &= 73442.6 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{\text{xs}} \cdot (LR + t_n)/2 \\ &= 194.759 \cdot 1542.799 \cdot (303.0 + 78.5)/2 \\ &= 573153.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{\text{xs}} \cdot R_{\text{nc}})/2 \\ &= (194.759 \cdot 1542.799 \cdot 151.5)/2 \\ &= 227608.8 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (573154 + 227608.8 + 73442.6)/194.759 \\ &= 4488.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{\text{allow}} / (2 \cdot A_p / AT - R_{\text{xs}} / (2 \cdot t_{\text{eff}})) \\ &= 2520.636 / (2 \cdot 4488.655 / 435.179 - 1542.799 / (2 \cdot 91.5)) \\ &= 206.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / [t / R_{\text{xs}}] \\ &= 2 \cdot 1680.424 \cdot 91.5 / 1542.799 \\ &= 199.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{\text{max1}}, P_{\text{max2}}) \\ &= \min(206.637, 199.324) \\ &= 199.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (73442.586 + 573154 + 227608.797) / 435.179 \\ &= 2008.840 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 194.759 \cdot 1542.799 / (2 \cdot 91.5) \\ &= 1641.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 2008.84 - 1641.935, 1641.935) \\ &= 2375.7 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2375.75	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	199.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (151.5 + 78.5) / 151.5 \\ &= 1.518 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2} \\ &= \pi/2 \cdot \sqrt{((151.5 + 78.5)^2 + (151.5 + 78.5)^2) / 2} \\ &= 361.283 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(227609 \cdot 1.52, 1.5 \cdot 1680.4(157.435 + 0.0), \pi/4 \cdot 194.8 \cdot 151.5^2 \cdot 1.52^2) \\ &= 80917.547 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 80917.547 / (361.283(0.49 \cdot 7.071 + 0.6 \cdot 91.5 + 0.49 \cdot 0.0)) \\ &= 383.747 < \text{ or } = \text{ to } 1680.424 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 17.7862 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 610.7863 mm

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Input, Nozzle Desc: A1 12"

From: 10

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Hemispherical Head	D	2992.00	mm
Head Finished (Minimum) Thickness	t	93.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1000.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

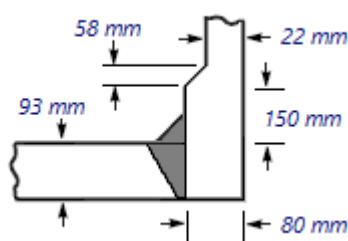
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		300.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	58.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	93.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		45.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: A1 12"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	300.000	mm.
Actual Thickness Used in Calculation	22.000	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 303.0 \cdot (\exp(194.759/(1680.42 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 18.6167 + 1.5 + 0.0 = 20.1167 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.42 \cdot 1.0 \cdot \ln((2 \cdot 20.5 + 303.0) / 303.0) - 0.0$$

$$= 213.26 - 0.0 = 213.2601 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
 Min. required shell thickness, internal p + c : 90.869 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 21.996 mm , is greater than required thickness 9.840 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 22.0 - 1.5$$

$$= 20.500 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2992.0/2 + 1.5$$

$$= 1497.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1497.5 \cdot 91.5}, 2 \cdot 151.5)$$

$$= 303.000 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((91.5 + 0.0)/78.5)^{0.35}, 1)$$

$$= \min(1.055, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + t_e + F_p \cdot \sqrt{R_n \cdot t_n}, L_{pr1} + t)$$

$$= \min(91.5 + 0.0 + 1.0 \cdot \sqrt{151.5 \cdot 78.5}, 500.0 + 91.5)$$

$$= 200.554 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 91.5 + ((0.0 \cdot 1.0) / 303.0)$$

$$= 91.500 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :



Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 91.5 \cdot 303.0 \\ &= 277.245 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 78.5 \cdot 200.554 \\ &= 157.435 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 277.245 + 1 \cdot (157.435 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 435.179 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 78.5 / \ln[1 + 78.5/151.5] \\ &= 188.027 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{\text{eff}} / \ln[1 + t_{\text{eff}}/R_{\text{eff}}] \\ &= 91.5 / \ln[1 + 91.5/1497.5] \\ &= 1542.799 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 188.027 \cdot 200.554 \\ &= 73442.6 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\ &= 194.759 \cdot 1542.799 \cdot (303.0 + 78.5)/2 \\ &= 573153.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (194.759 \cdot 1542.799 \cdot 151.5)/2 \\ &= 227608.8 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (573154 + 227608.8 + 73442.6)/194.759 \\ &= 4488.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{\text{allow}} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{\text{eff}})) \\ &= 2520.636 / (2 \cdot 4488.655 / 435.179 - 1542.799 / (2 \cdot 91.5)) \\ &= 206.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / [t / R_{xs}] \\ &= 2 \cdot 1680.424 \cdot [91.5 / 1542.799] \\ &= 199.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{\text{max1}}, P_{\text{max2}}) \\ &= \min(206.637, 199.324) \\ &= 199.324 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (73442.586 + 573154 + 227608.797) / 435.179 \\ &= 2008.840 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 194.759 \cdot 1542.799 / (2 \cdot 91.5) \\ &= 1641.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 2008.84 - 1641.935, 1641.935) \\ &= 2375.7 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2375.75	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	199.32	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (R_{nc} + t_n) / R_{nc} \\ &= (151.5 + 78.5) / 151.5 \\ &= 1.518 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc} + t_n)^2 + (R_n + t_n)^2) / 2} \\ &= \pi/2 \cdot \sqrt{((151.5 + 78.5)^2 + (151.5 + 78.5)^2) / 2} \\ &= 361.283 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(227609 \cdot 1.52, 1.5 \cdot 1680.4(157.435 + 0.0), \pi/4 \cdot 194.8 \cdot 151.5^2 \cdot 1.52^2) \\ &= 80917.547 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T)) \\ &= 80917.547 / (361.283(0.49 \cdot 7.071 + 0.6 \cdot 91.5 + 0.49 \cdot 0.0)) \\ &= 383.747 < \text{ or } = \text{ to } 1680.424 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 17.7862 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 610.7863 mm

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Input, Nozzle Desc: T11 2"

From: 40

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		455.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

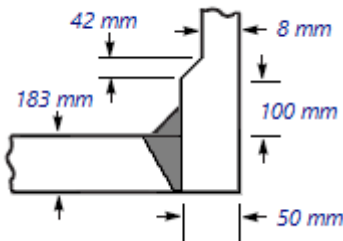
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T11 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 194.759 \cdot 1540.469(53.0 + 48.5) \\
 &= 304520.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 194.759 \cdot 1540.469 \cdot 26.5 \\
 &= 79505.2 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (304520.1 + 79505.2 + 19734.3)/194.759 \\
 &= 2073.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\
 &= 209.6 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 1680.424[181.5/1540.469] \\
 &= 198.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(209.574, 197.99) \\
 &= 197.990 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (19734.334 + 304520.062 + 79505.234)/202.11 \\
 &= 1997.726 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 194.759 \cdot 1540.469/181.5 \\
 &= 1653.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\
 &= 2342.4 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (26.5 + 48.5)/26.5 \\
 &= 2.830 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (26.5 + 48.5) \\
 &= 117.810 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415 + 0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2)
 \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0))$
= 64.998 < or = to 1680.424 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-4 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	21 °C
Temperature reduction per Graph 3.12	:	24 °C
The computed stress ratio is	:	0.394

The computed thickness ratio [sr]:

= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(2.562 \cdot 1.0) / (8.0 - 1.5 - 0.0)$
= 0.394

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T12 2"

From: 40

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		455.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

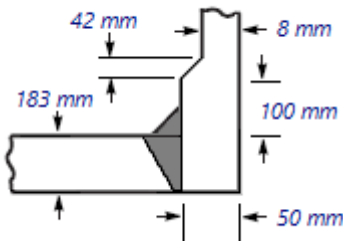
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T12 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 128.00 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 181.5, 0.0] + \sqrt{(26.5 \cdot 48.5)} \\ &= 35.850 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \frac{(2 \cdot R_n + t_n) / \sqrt{(D_i + t_{eff}) \cdot t_{eff}}}{(2 \cdot 26.5 + 48.5) / \sqrt{(2903.0 + 181.5) \cdot 181.5}}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 181.5 \cdot 53.0 \cdot \max((0.136/5)^{0.85}, 1) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot LH$$



$$= 194.759 \cdot 46.619 \cdot 217.35$$
$$= 19734.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$
$$= 194.759 \cdot 1540.469(53.0 + 48.5)$$
$$= 304520.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 194.759 \cdot 1540.469 \cdot 26.5$$
$$= 79505.2 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (304520.1 + 79505.2 + 19734.3)/194.759$$
$$= 2073.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5)$$
$$= 209.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1680.424[181.5/1540.469]$$
$$= 198.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(209.574, 197.99)$$
$$= 197.990 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (19734.334 + 304520.062 + 79505.234)/202.11$$
$$= 1997.726 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$
$$= 194.759 \cdot 1540.469/181.5$$
$$= 1653.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$
$$= \max(2 \cdot 1997.726 - 1653.004, 1653.004)$$
$$= 2342.4 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (26.5 + 48.5)/26.5$$
$$= 2.830 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (26.5 + 48.5)$$
$$= 117.810 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$



Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415 + 0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2) \\ &= 8604.181 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0)) \\ &= 64.998 < \text{or} = \text{to } 1680.424 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -4 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.7M- Non PWHT, the MDMT is	: 21 °C
Temperature reduction per Graph 3.12	: 24 °C
The computed stress ratio is	: 0.394

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (2.562 \cdot 1.0) / (8.0 - 1.5 - 0.0) \\ &= 0.394 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T13 2"

From: 40

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		455.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

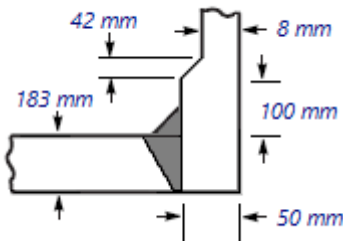
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T13 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 194.759 \cdot 1540.469(53.0 + 48.5) \\ &= 304520.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 194.759 \cdot 1540.469 \cdot 26.5 \\ &= 79505.2 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (304520.1 + 79505.2 + 19734.3)/194.759 \\ &= 2073.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\ &= 209.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1680.424[181.5/1540.469] \\ &= 198.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(209.574, 197.99) \\ &= 197.990 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (19734.334 + 304520.062 + 79505.234)/202.11 \\ &= 1997.726 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 194.759 \cdot 1540.469/181.5 \\ &= 1653.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\ &= 2342.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (26.5 + 48.5)/26.5 \\ &= 2.830 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (26.5 + 48.5) \\ &= 117.810 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415+0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2) \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0))$
= 64.998 < or = to 1680.424 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-4 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	21 °C
Temperature reduction per Graph 3.12	:	24 °C
The computed stress ratio is	:	0.394

The computed thickness ratio [sr]:

= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(2.562 \cdot 1.0) / (8.0 - 1.5 - 0.0)$
= 0.394

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T8 2"

From: 70

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1450.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

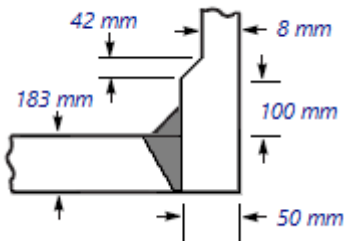
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T8 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 194.759 \cdot 1540.469(53.0 + 48.5) \\
 &= 304520.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 194.759 \cdot 1540.469 \cdot 26.5 \\
 &= 79505.2 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (304520.1 + 79505.2 + 19734.3)/194.759 \\
 &= 2073.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\
 &= 209.6 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 1680.424[181.5/1540.469] \\
 &= 198.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(209.574, 197.99) \\
 &= 197.990 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (19734.334 + 304520.062 + 79505.234)/202.11 \\
 &= 1997.726 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 194.759 \cdot 1540.469/181.5 \\
 &= 1653.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\
 &= 2342.4 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (26.5 + 48.5)/26.5 \\
 &= 2.830 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (26.5 + 48.5) \\
 &= 117.810 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415+0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2)
 \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0))$
= 64.998 < or = to 1680.424 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-4 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	21 °C
Temperature reduction per Graph 3.12	:	24 °C
The computed stress ratio is	:	0.394

The computed thickness ratio [sr]:

= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(2.562 \cdot 1.0) / (8.0 - 1.5 - 0.0)$
= 0.394

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T9 2"

From: 70

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1450.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

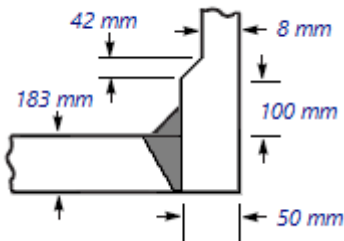
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T9 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 194.759 \cdot 1540.469(53.0 + 48.5) \\
 &= 304520.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 194.759 \cdot 1540.469 \cdot 26.5 \\
 &= 79505.2 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (304520.1 + 79505.2 + 19734.3)/194.759 \\
 &= 2073.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\
 &= 209.6 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 1680.424[181.5/1540.469] \\
 &= 198.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(209.574, 197.99) \\
 &= 197.990 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (19734.334 + 304520.062 + 79505.234)/202.11 \\
 &= 1997.726 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 194.759 \cdot 1540.469/181.5 \\
 &= 1653.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\
 &= 2342.4 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (26.5 + 48.5)/26.5 \\
 &= 2.830 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (26.5 + 48.5) \\
 &= 117.810 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415+0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2)
 \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0)) \\ &= 64.998 < \text{or} = \text{to } 1680.424 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-4 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	21 °C
Temperature reduction per Graph 3.12	:	24 °C
The computed stress ratio is	:	0.394

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (2.562 \cdot 1.0) / (8.0 - 1.5 - 0.0) \\ &= 0.394 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T10 2"

From: 70

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1450.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

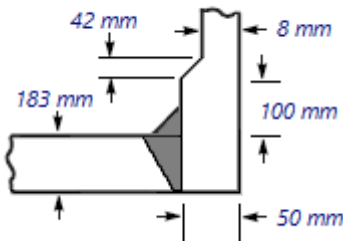
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T10 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 194.759 \cdot 1540.469(53.0 + 48.5) \\
 &= 304520.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 194.759 \cdot 1540.469 \cdot 26.5 \\
 &= 79505.2 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (304520.1 + 79505.2 + 19734.3)/194.759 \\
 &= 2073.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\
 &= 209.6 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 1680.424[181.5/1540.469] \\
 &= 198.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(209.574, 197.99) \\
 &= 197.990 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (19734.334 + 304520.062 + 79505.234)/202.11 \\
 &= 1997.726 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 194.759 \cdot 1540.469/181.5 \\
 &= 1653.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\
 &= 2342.4 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (26.5 + 48.5)/26.5 \\
 &= 2.830 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (26.5 + 48.5) \\
 &= 117.810 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415 + 0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2)
 \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0))$
= $64.998 < \text{or} = \text{to } 1680.424$ Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-4 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	21 °C
Temperature reduction per Graph 3.12	:	24 °C
The computed stress ratio is	:	0.394

The computed thickness ratio [sr]:

= $(t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
= $(2.562 \cdot 1.0) / (8.0 - 1.5 - 0.0)$
= 0.394

Warning: Impact Testing Is Mandatory!

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -4 °C

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T5 2"

From: 110

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2600.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

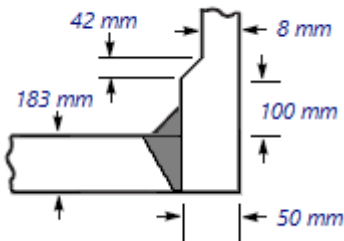
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T5 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 128.00 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 181.5, 0.0] + \sqrt{(26.5 \cdot 48.5)} \\ &= 35.850 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \frac{(2 \cdot R_n + t_n) / \sqrt{(D_i + teff) \cdot teff}}{1}\right) \\ &= \min\left(12, \frac{(2 \cdot 26.5 + 48.5) / \sqrt{(2903.0 + 181.5) \cdot 181.5}}{1}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 181.5 \cdot 53.0 \cdot \max((0.136/5)^{0.85}, 1) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg_1^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$



$$= 194.759 \cdot 46.619 \cdot 217.35$$
$$= 19734.3 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$
$$= 194.759 \cdot 1540.469(53.0 + 48.5)$$
$$= 304520.1 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 194.759 \cdot 1540.469 \cdot 26.5$$
$$= 79505.2 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (304520.1 + 79505.2 + 19734.3)/194.759$$
$$= 2073.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5)$$
$$= 209.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1680.424[181.5/1540.469]$$
$$= 198.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(209.574, 197.99)$$
$$= 197.990 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ Avg]:

$$= (fN + fS + fY) / AT$$
$$= (19734.334 + 304520.062 + 79505.234)/202.11$$
$$= 1997.726 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ Circ]:

$$= P \cdot Rxs / teff$$
$$= 194.759 \cdot 1540.469/181.5$$
$$= 1653.0 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$
$$= \max(2 \cdot 1997.726 - 1653.004, 1653.004)$$
$$= 2342.4 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (26.5 + 48.5)/26.5$$
$$= 2.830 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$
$$= \pi/2 \cdot (26.5 + 48.5)$$
$$= 117.810 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415 + 0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2) \\ &= 8604.181 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0)) \\ &= 64.998 < \text{or} = \text{to } 1680.424 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T6 2"

From: 110

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2600.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

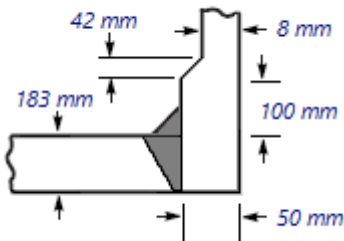
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T6 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned}
 &= P \cdot Rxs(LR + tn) \\
 &= 194.759 \cdot 1540.469(53.0 + 48.5) \\
 &= 304520.1 \text{ kgf}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot Rxs \cdot Rnc \\
 &= 194.759 \cdot 1540.469 \cdot 26.5 \\
 &= 79505.2 \text{ kgf}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (fS + fY + fN)/P \\
 &= (304520.1 + 79505.2 + 19734.3)/194.759 \\
 &= 2073.1 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\
 &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\
 &= 209.6 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S[t/Rxs] \\
 &= 1680.424[181.5/1540.469] \\
 &= 198.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(Pmax1, Pmax2) \\
 &= \min(209.574, 197.99) \\
 &= 197.990 \text{ kgf/cm}^2
 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned}
 &= (fN + fS + fY) / AT \\
 &= (19734.334 + 304520.062 + 79505.234)/202.11 \\
 &= 1997.726 \text{ kgf/cm}^2
 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned}
 &= P \cdot Rxs / teff \\
 &= 194.759 \cdot 1540.469/181.5 \\
 &= 1653.0 \text{ kgf/cm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\
 &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\
 &= 2342.4 \text{ kgf/cm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (Rnc + tn) / Rnc \\
 &= (26.5 + 48.5)/26.5 \\
 &= 2.830 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned}
 &= \pi/2 \cdot (Rn + tn) \\
 &= \pi/2 \cdot (26.5 + 48.5) \\
 &= 117.810 \text{ mm}
 \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned}
 &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\
 &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415 + 0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2)
 \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [τ]:

= $f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$

= $8604.181 / (117.81(0.49 \cdot 7.071 + 0.6 \cdot 181.5 + 0.49 \cdot 0.0))$

= 64.998 < or = to 1680.424 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T7 2"

From: 110

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-387 11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2900.00	mm
Shell Finished (Minimum) Thickness	t	183.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2600.00	cm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

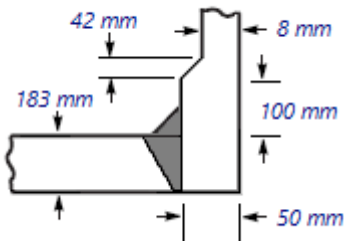
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	+IT
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	42.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	183.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T7 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0$$

$$= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 6.020 mm
 Min. required shell thickness, internal p + c : 180.075 mm

Nozzle neck minimum required thickness : 6.020 mm

Minimum nozzle neck thickness 8.001 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1680.4/1680.4, 1]$$

$$= 1.000$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.0 - 1.5$$

$$= 6.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2903.0/181.5$$

$$= 15.994 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2900.0/2 + 1.5$$

$$= 1451.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1451.5 \cdot 181.5}, 2 \cdot 26.5)$$

$$= 53.000 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 181.5, 0.0] + \sqrt{26.5 \cdot 48.5}$$



$$= 35.850 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 300.0 \\ &= 300.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(181.5 + 0.0) \\ &= 1452.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[35.85, 300.0, 1452.0] + 181.5 \\ &= 217.350 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 181.5 + ((0.0 \cdot 1.0) / 53.0) \\ &= 181.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min\left(12, \left(2 \cdot R_n + t_n\right) / \sqrt{\left(\left(D_i + t_{eff}\right) \cdot t_{eff}\right)}\right) \\ &= \min\left(12, \left(2 \cdot 26.5 + 48.5\right) / \sqrt{\left(\left(2903.0 + 181.5\right) \cdot 181.5\right)}\right) \\ &= 0.136 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max\left(\left(\lambda/5\right)^{0.85}, 1\right) \\ &= 181.5 \cdot 53.0 \cdot \max\left(\left(0.136/5\right)^{0.85}, 1\right) \\ &= 96.195 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 48.5 \cdot 217.35 \\ &= 105.415 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 96.195 + 1.0(105.415 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 202.110 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 48.5 / \ln[1 + 48.5/26.5] \\ &= 46.619 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 181.5 / \ln[1 + 181.5/1451.5] \\ &= 1540.469 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1680.424 \cdot 1.0 \\ &= 2520.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot LH \\ &= 194.759 \cdot 46.619 \cdot 217.35 \\ &= 19734.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:



$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 194.759 \cdot 1540.469(53.0 + 48.5) \\ &= 304520.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 194.759 \cdot 1540.469 \cdot 26.5 \\ &= 79505.2 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (304520.1 + 79505.2 + 19734.3)/194.759 \\ &= 2073.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2520.636 / (2 \cdot 2073.127/202.11 - 1540.469/181.5) \\ &= 209.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1680.424[181.5/1540.469] \\ &= 198.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(209.574, 197.99) \\ &= 197.990 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (19734.334 + 304520.062 + 79505.234)/202.11 \\ &= 1997.726 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 194.759 \cdot 1540.469/181.5 \\ &= 1653.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1997.726 - 1653.004, 1653.004) \\ &= 2342.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	2342.45	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	197.99	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (26.5 + 48.5)/26.5 \\ &= 2.830 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (26.5 + 48.5) \\ &= 117.810 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(79505 \cdot 2.83, 1.5 \cdot 1680.4(105.415+0.0), \pi/4 \cdot 194.8 \cdot 26.5^2 \cdot 2.83^2) \end{aligned}$$



= 8604.181 kgf

Weld Stress Value [T]:

= fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))

= 8604.181/(117.81(0.49•7.071 + 0.6•181.5 + 0.49•0.0))

= 64.998 < or = to 1680.424 Weld Stress Passed

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

The Drop for this Nozzle is : 1.9410 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 484.9409 mm

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Input, Nozzle Desc: T1 2"

From: 130

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Outside Diameter of Bolted Blind Flange	D	4000.00	mm
Head Finished (Minimum) Thickness	t	700.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1300.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

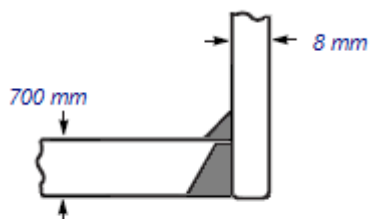
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	700.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T1 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0 \\ &= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.020 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 1.500 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 1.500 \text{ mm}$$

Minimum nozzle neck thickness 8.001 mm, is greater than required thickness 1.500 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1680.4/1680.4, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 8.0 - 1.5 \\ &= 6.500 \text{ mm} \end{aligned}$$

Nozzle Mean Radius [Rnm]:

$$\begin{aligned} &= R_n + 0.5 \cdot t_n \\ &= 26.5 + 0.5 \cdot 6.5 \\ &= 29.750 \text{ mm} \end{aligned}$$

Parameter [λN]:

$$\begin{aligned} &= 1.285 / \sqrt{(R_{nm} \cdot t_n)} \\ &= 1.285 / \sqrt{(29.75 \cdot 6.5)} \\ &= 0.092 \text{ 1/mm} \end{aligned}$$

Parameter [CL]:

$$\begin{aligned} &= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3) \\ &= \min(0.092 \cdot (300.0 + 698.5 + 0.0), 3) \\ &= 3.000 \end{aligned}$$

Parameter [C1]:

$$\begin{aligned} &= \sinh(CL)^2 + \sin(CL)^2 \\ &= 100.358 + 0.02 \\ &= 100.378 \end{aligned}$$

Parameter [C2]:

$$\begin{aligned} &= \sinh(CL)^2 - \sin(CL)^2 \\ &= 100.358 - 0.02 \\ &= 100.338 \end{aligned}$$

Parameter [C3]:

$$\begin{aligned} &= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2})) \\ &= (300.0 + 698.5) / (300.0 + 698.5 + \min(1/2.347, 0.0)) \\ &= 1.000 \end{aligned}$$

Parameter [Rxn]:

$$\begin{aligned} &= t_n / \ln((R_n + t_n) / R_n) \\ &= 6.5 / \ln((26.5 + 6.5) / 26.5) \\ &= 29.631 \text{ mm} \end{aligned}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N \cdot (t + t_e + L_{41} + L_{43})$$



$$= 0.5 \cdot 0.092 \cdot (698.5 + 0.0 + 10.0 + 0.0)$$
$$= 32.735$$

Parameter [Ct]:

$$= \exp(-xt)$$
$$= \exp(-32.735)$$
$$= 0.000$$

Parameter [Cp]:

$$= \min((W + 0.5 \cdot L42) / (Rn \cdot t))te, 0.6)$$
$$= \min((0.0 + 0.5 \cdot 0.0) / (26.5 \cdot 698.5)0.0, 0.6)$$
$$= 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot trf^4 / (6 \cdot (t + Cp \cdot te)^2)$$
$$= 1680.424 \cdot 693.976^4 / (6 \cdot (698.5 + 0.0 \cdot 0.0)^2)$$
$$= 1331409.500 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot Mo \cdot \lambda N^2 \cdot Rnm \cdot (CT \cdot C1 \cdot C3) / (tn \cdot C2) + P \cdot Rxn / tn$$
$$= 2 \cdot 1331410 \cdot 0.092^2 \cdot 29.75 \cdot (0.0) / (6.5 \cdot 100.338) +$$
$$194.759 \cdot 29.631 / 6.5$$
$$= 887.840 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	887.84	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$
$$= (26.5 + 6.5) / 26.5$$
$$= 1.245 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$
$$= \pi / 2 \cdot (26.5 + 6.5)$$
$$= 51.836 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 6.500, 6.000)$ Passed

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot trf^4 / t^3$$
$$= 0.3 \cdot 1680.424 \cdot 693.976^4 / 698.5^3$$
$$= 3430.977 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(tw1, 0.5 \cdot t)$$
$$= \min(698.5, 0.5 \cdot 698.5)$$
$$= 349.250 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((tw1 - 0.5 \cdot t), 0.0), 0.5 \cdot t)$$
$$= \min(\max((698.5 - 0.5 \cdot 698.5), 0.0), 0.5 \cdot 698.5)$$
$$= 349.250 \text{ mm}$$

Shear Stress [τ1]:

$$= Vs / (0.6 \cdot tx1 + 0.49 \cdot L43T)$$
$$= 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 0.0)$$
$$= 1637.309 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= Vs / (0.6 \cdot tx2 + 0.49 \cdot L41T)$$
$$= 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 7.071)$$
$$= 1610.678 \text{ kgf/cm}^2$$



Shear Stress [τ_3]:

$$\begin{aligned} &= P/2(R_n + t_n)/(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot R_{L43T}) \\ &= 194.76/2 \cdot (26.5 + 6.5)/(0.49 \cdot 7.071 + 0.6 \cdot 698.5 + 0.49 \cdot 0.0) \\ &= 7.605 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1637.3, 1610.7, 7.6) \\ &= 1637.3 \text{ must be less than or equal to } 1680.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -25 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: -1 °C
Temperature reduction per Graph 3.12	: 24 °C
The computed stress ratio is	: 0.394

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*)/(t_{nom} - c_i - c_{ext}) \\ &= (2.562 \cdot 1.0)/(8.0 - 1.5 - 0.0) \\ &= 0.394 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -25 °C

The Drop for this Nozzle is : 0.0508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1000.0508 mm

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Input, Nozzle Desc: T2 2"

From: 130

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Outside Diameter of Bolted Blind Flange	D	4000.00	mm
Head Finished (Minimum) Thickness	t	700.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1300.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

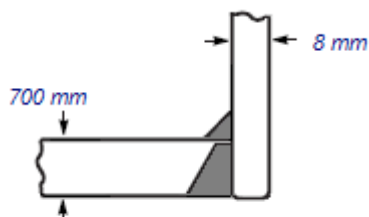
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	700.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T2 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0 \\ &= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.020 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 1.500 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 1.500 \text{ mm}$$

Minimum nozzle neck thickness 8.001 mm, is greater than required thickness 1.500 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1680.4/1680.4, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 8.0 - 1.5 \\ &= 6.500 \text{ mm} \end{aligned}$$

Nozzle Mean Radius [Rnm]:

$$\begin{aligned} &= R_n + 0.5 \cdot t_n \\ &= 26.5 + 0.5 \cdot 6.5 \\ &= 29.750 \text{ mm} \end{aligned}$$

Parameter [λN]:

$$\begin{aligned} &= 1.285 / \sqrt{(R_{nm} \cdot t_n)} \\ &= 1.285 / \sqrt{(29.75 \cdot 6.5)} \\ &= 0.092 \text{ 1/mm} \end{aligned}$$

Parameter [CL]:

$$\begin{aligned} &= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3) \\ &= \min(0.092 \cdot (300.0 + 698.5 + 0.0), 3) \\ &= 3.000 \end{aligned}$$

Parameter [C1]:

$$\begin{aligned} &= \sinh(CL)^2 + \sin(CL)^2 \\ &= 100.358 + 0.02 \\ &= 100.378 \end{aligned}$$

Parameter [C2]:

$$\begin{aligned} &= \sinh(CL)^2 - \sin(CL)^2 \\ &= 100.358 - 0.02 \\ &= 100.338 \end{aligned}$$

Parameter [C3]:

$$\begin{aligned} &= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2})) \\ &= (300.0 + 698.5) / (300.0 + 698.5 + \min(1/2.347, 0.0)) \\ &= 1.000 \end{aligned}$$

Parameter [Rxn]:

$$\begin{aligned} &= t_n / \ln((R_n + t_n) / R_n) \\ &= 6.5 / \ln((26.5 + 6.5) / 26.5) \\ &= 29.631 \text{ mm} \end{aligned}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N \cdot (t + t_e + L_{41} + L_{43})$$



$$= 0.5 \cdot 0.092 \cdot (698.5 + 0.0 + 10.0 + 0.0) \\ = 32.735$$

Parameter [Ct]:

$$= \exp(-xt) \\ = \exp(-32.735) \\ = 0.000$$

Parameter [Cp]:

$$= \min((W + 0.5 \cdot L42) / (Rn \cdot t))te, 0.6) \\ = \min((0.0 + 0.5 \cdot 0.0) / (26.5 \cdot 698.5)0.0, 0.6) \\ = 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot trf^4 / (6 \cdot (t + Cp \cdot te)^2) \\ = 1680.424 \cdot 693.976^4 / (6 \cdot (698.5 + 0.0 \cdot 0.0)^2) \\ = 1331409.500 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot Mo \cdot \lambda N^2 \cdot Rnm \cdot (CT \cdot C1 \cdot C3) / (tn \cdot C2) + P \cdot Rxn / tn \\ = 2 \cdot 1331410 \cdot 0.092^2 \cdot 29.75 \cdot (0.0) / (6.5 \cdot 100.338) + \\ 194.759 \cdot 29.631 / 6.5 \\ = 887.840 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	887.84	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc \\ = (26.5 + 6.5) / 26.5 \\ = 1.245 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn) \\ = \pi / 2 \cdot (26.5 + 6.5) \\ = 51.836 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 6.500, 6.000)$ Passed

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot trf^4 / t^3 \\ = 0.3 \cdot 1680.424 \cdot 693.976^4 / 698.5^3 \\ = 3430.977 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(tw1, 0.5 \cdot t) \\ = \min(698.5, 0.5 \cdot 698.5) \\ = 349.250 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((tw1 - 0.5 \cdot t), 0.0), 0.5 \cdot t) \\ = \min(\max((698.5 - 0.5 \cdot 698.5), 0.0), 0.5 \cdot 698.5) \\ = 349.250 \text{ mm}$$

Shear Stress [τ1]:

$$= Vs / (0.6 \cdot tx1 + 0.49 \cdot L43T) \\ = 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 0.0) \\ = 1637.309 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= Vs / (0.6 \cdot tx2 + 0.49 \cdot L41T) \\ = 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 7.071) \\ = 1610.678 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$\begin{aligned} &= P/2(R_n + t_n)/(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot R_{L43T}) \\ &= 194.76/2 \cdot (26.5 + 6.5)/(0.49 \cdot 7.071 + 0.6 \cdot 698.5 + 0.49 \cdot 0.0) \\ &= 7.605 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1637.3, 1610.7, 7.6) \\ &= 1637.3 \text{ must be less than or equal to } 1680.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -25 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: -1 °C
Temperature reduction per Graph 3.12	: 24 °C
The computed stress ratio is	: 0.394

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*)/(t_{nom} - c_i - c_{ext}) \\ &= (2.562 \cdot 1.0)/(8.0 - 1.5 - 0.0) \\ &= 0.394 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -25 °C

The Drop for this Nozzle is : 0.0508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1000.0508 mm

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Input, Nozzle Desc: T3 2"

From: 130

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Outside Diameter of Bolted Blind Flange	D	4000.00	mm
Head Finished (Minimum) Thickness	t	700.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1300.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

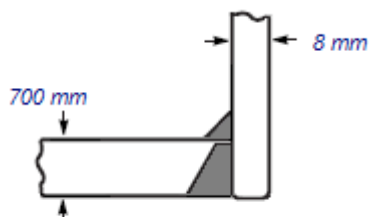
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	700.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T3 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0 \\ &= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.020 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 1.500 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 1.500 \text{ mm}$$

Minimum nozzle neck thickness 8.001 mm, is greater than required thickness 1.500 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1680.4/1680.4, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 8.0 - 1.5 \\ &= 6.500 \text{ mm} \end{aligned}$$

Nozzle Mean Radius [Rnm]:

$$\begin{aligned} &= R_n + 0.5 \cdot t_n \\ &= 26.5 + 0.5 \cdot 6.5 \\ &= 29.750 \text{ mm} \end{aligned}$$

Parameter [λN]:

$$\begin{aligned} &= 1.285 / \sqrt{(R_{nm} \cdot t_n)} \\ &= 1.285 / \sqrt{(29.75 \cdot 6.5)} \\ &= 0.092 \text{ 1/mm} \end{aligned}$$

Parameter [CL]:

$$\begin{aligned} &= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3) \\ &= \min(0.092 \cdot (300.0 + 698.5 + 0.0), 3) \\ &= 3.000 \end{aligned}$$

Parameter [C1]:

$$\begin{aligned} &= \sinh(CL)^2 + \sin(CL)^2 \\ &= 100.358 + 0.02 \\ &= 100.378 \end{aligned}$$

Parameter [C2]:

$$\begin{aligned} &= \sinh(CL)^2 - \sin(CL)^2 \\ &= 100.358 - 0.02 \\ &= 100.338 \end{aligned}$$

Parameter [C3]:

$$\begin{aligned} &= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2})) \\ &= (300.0 + 698.5) / (300.0 + 698.5 + \min(1/2.347, 0.0)) \\ &= 1.000 \end{aligned}$$

Parameter [Rxn]:

$$\begin{aligned} &= t_n / \ln((R_n + t_n) / R_n) \\ &= 6.5 / \ln((26.5 + 6.5) / 26.5) \\ &= 29.631 \text{ mm} \end{aligned}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N \cdot (t + t_e + L_{41} + L_{43})$$



$$= 0.5 \cdot 0.092 \cdot (698.5 + 0.0 + 10.0 + 0.0) \\ = 32.735$$

Parameter [Ct]:

$$= \exp(-xt) \\ = \exp(-32.735) \\ = 0.000$$

Parameter [Cp]:

$$= \min((W + 0.5 \cdot L42) / (Rn \cdot t))te, 0.6) \\ = \min((0.0 + 0.5 \cdot 0.0) / (26.5 \cdot 698.5)0.0, 0.6) \\ = 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot trf^4 / (6 \cdot (t + Cp \cdot te)^2) \\ = 1680.424 \cdot 693.976^4 / (6 \cdot (698.5 + 0.0 \cdot 0.0)^2) \\ = 1331409.500 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot Mo \cdot \lambda N^2 \cdot Rnm \cdot (CT \cdot C1 \cdot C3) / (tn \cdot C2) + P \cdot Rxn / tn \\ = 2 \cdot 1331410 \cdot 0.092^2 \cdot 29.75 \cdot (0.0) / (6.5 \cdot 100.338) + \\ 194.759 \cdot 29.631 / 6.5 \\ = 887.840 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	887.84	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc \\ = (26.5 + 6.5) / 26.5 \\ = 1.245 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn) \\ = \pi / 2 \cdot (26.5 + 6.5) \\ = 51.836 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 6.500, 6.000)$ Passed

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot trf^4 / t^3 \\ = 0.3 \cdot 1680.424 \cdot 693.976^4 / 698.5^3 \\ = 3430.977 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(tw1, 0.5 \cdot t) \\ = \min(698.5, 0.5 \cdot 698.5) \\ = 349.250 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((tw1 - 0.5 \cdot t), 0.0), 0.5 \cdot t) \\ = \min(\max((698.5 - 0.5 \cdot 698.5), 0.0), 0.5 \cdot 698.5) \\ = 349.250 \text{ mm}$$

Shear Stress [τ1]:

$$= Vs / (0.6 \cdot tx1 + 0.49 \cdot L43T) \\ = 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 0.0) \\ = 1637.309 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= Vs / (0.6 \cdot tx2 + 0.49 \cdot L41T) \\ = 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 7.071) \\ = 1610.678 \text{ kgf/cm}^2$$



Shear Stress [τ_3]:

$$\begin{aligned} &= P/2(R_n + t_n)/(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot R_{L43T}) \\ &= 194.76/2 \cdot (26.5 + 6.5)/(0.49 \cdot 7.071 + 0.6 \cdot 698.5 + 0.49 \cdot 0.0) \\ &= 7.605 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1637.3, 1610.7, 7.6) \\ &= 1637.3 \text{ must be less than or equal to } 1680.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -25 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: -1 °C
Temperature reduction per Graph 3.12	: 24 °C
The computed stress ratio is	: 0.394

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*)/(t_{nom} - c_i - c_{ext}) \\ &= (2.562 \cdot 1.0)/(8.0 - 1.5 - 0.0) \\ &= 0.394 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -25 °C

The Drop for this Nozzle is : 0.0508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1000.0508 mm

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Input, Nozzle Desc: T4 2"

From: 130

Pressure for Reinforcement Calculations	P	194.759	kgf/cm ²
Temperature for Internal Pressure	Temp	290	°C
Parent Material		SA-336 F11	+IT
Parent Allowable Stress at Temperature	Sv	1680.42	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	2110.70	kgf/cm ²
Outside Diameter of Bolted Blind Flange	D	4000.00	mm
Head Finished (Minimum) Thickness	t	700.0000	mm
Head Internal Corrosion Allowance	cas	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1300.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

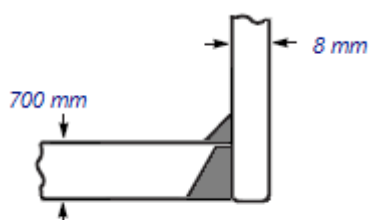
Material		SA-336 F11	+IT
Material UNS Number		K11572	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1680.42	kgf/cm ²
Allowable Stress At Ambient	Sna	2110.70	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		50.0000	mm
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm
Flange Material		SA-336 F11	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	700.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T4 2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	50.000	mm.
Actual Thickness Used in Calculation	8.000	mm



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 53.0 \cdot (\exp(194.759/(1680.424 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 3.2564 + 1.5 + 0.0 = 4.7564 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1680.424 \cdot 1.0 \cdot \ln((2 \cdot 6.5 + 53.0) / 53.0) - 0.0 \\ &= 368.623 - 0.0 = 368.6226 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

$$\begin{aligned} \text{Min. required nozzle neck thickness, table 4.5.2 + c} &: 6.020 \text{ mm} \\ \text{Min. required shell thickness, internal p + c} &: 1.500 \text{ mm} \end{aligned}$$

$$\text{Nozzle neck minimum required thickness} : 1.500 \text{ mm}$$

Minimum nozzle neck thickness 8.001 mm, is greater than required thickness 1.500 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1680.4/1680.4, 1] \\ &= 1.000 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 8.0 - 1.5 \\ &= 6.500 \text{ mm} \end{aligned}$$

Nozzle Mean Radius [Rnm]:

$$\begin{aligned} &= R_n + 0.5 \cdot t_n \\ &= 26.5 + 0.5 \cdot 6.5 \\ &= 29.750 \text{ mm} \end{aligned}$$

Parameter [λN]:

$$\begin{aligned} &= 1.285 / \sqrt{(R_{nm} \cdot t_n)} \\ &= 1.285 / \sqrt{(29.75 \cdot 6.5)} \\ &= 0.092 \text{ 1/mm} \end{aligned}$$

Parameter [CL]:

$$\begin{aligned} &= \min(\lambda N \cdot (L_{pr1} + t + L_{pr2}), 3) \\ &= \min(0.092 \cdot (300.0 + 698.5 + 0.0), 3) \\ &= 3.000 \end{aligned}$$

Parameter [C1]:

$$\begin{aligned} &= \sinh(CL)^2 + \sin(CL)^2 \\ &= 100.358 + 0.02 \\ &= 100.378 \end{aligned}$$

Parameter [C2]:

$$\begin{aligned} &= \sinh(CL)^2 - \sin(CL)^2 \\ &= 100.358 - 0.02 \\ &= 100.338 \end{aligned}$$

Parameter [C3]:

$$\begin{aligned} &= (L_{pr1} + t) / (L_{pr1} + t + \min(1/\lambda N, L_{pr2})) \\ &= (300.0 + 698.5) / (300.0 + 698.5 + \min(1/2.347, 0.0)) \\ &= 1.000 \end{aligned}$$

Parameter [Rxn]:

$$\begin{aligned} &= t_n / \ln((R_n + t_n) / R_n) \\ &= 6.5 / \ln((26.5 + 6.5) / 26.5) \\ &= 29.631 \text{ mm} \end{aligned}$$

Parameter [xt]:

$$= 0.5 \cdot \lambda N \cdot (t + t_e + L_{41} + L_{43})$$



$$= 0.5 \cdot 0.092 \cdot (698.5 + 0.0 + 10.0 + 0.0)$$

$$= 32.735$$

Parameter [Ct]:

$$= \exp(-xt)$$

$$= \exp(-32.735)$$

$$= 0.000$$

Parameter [Cp]:

$$= \min((W + 0.5 \cdot L42) / (Rn \cdot t))te, 0.6)$$

$$= \min((0.0 + 0.5 \cdot 0.0) / (26.5 \cdot 698.5)0.0, 0.6)$$

$$= 0.000$$

Maximum Unit Moment at the Nozzle Intersection [Mo]:

$$= S \cdot trf^4 / (6 \cdot (t + Cp \cdot te)^2)$$

$$= 1680.424 \cdot 693.976^4 / (6 \cdot (698.5 + 0.0 \cdot 0.0)^2)$$

$$= 1331409.500 \text{ kgf}$$

Maximum Local Primary Membrane Stress in the Nozzle [PL]:

$$= 2 \cdot Mo \cdot \lambda N^2 \cdot Rnm \cdot (CT \cdot C1 \cdot C3) / (tn \cdot C2) + P \cdot Rxn / tn$$

$$= 2 \cdot 1331410 \cdot 0.092^2 \cdot 29.75 \cdot (0.0) / (6.5 \cdot 100.338) +$$

$$194.759 \cdot 29.631 / 6.5$$

$$= 887.840 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2520.64	kgf/cm ²
Local Primary Membrane Stress	PL	887.84	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (26.5 + 6.5) / 26.5$$

$$= 1.245 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (26.5 + 6.5)$$

$$= 51.836 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 6.500, 6.000) \text{ Passed}$$

Unit Shear Force [Vs]:

$$= 0.3 \cdot S \cdot trf^4 / t^3$$

$$= 0.3 \cdot 1680.424 \cdot 693.976^4 / 698.5^3$$

$$= 3430.977 \text{ kgf/mm}$$

Thickness Value [tx1]:

$$= \min(tw1, 0.5 \cdot t)$$

$$= \min(698.5, 0.5 \cdot 698.5)$$

$$= 349.250 \text{ mm}$$

Thickness Value [tx2]:

$$= \min(\max((tw1 - 0.5 \cdot t), 0.0), 0.5 \cdot t)$$

$$= \min(\max((698.5 - 0.5 \cdot 698.5), 0.0), 0.5 \cdot 698.5)$$

$$= 349.250 \text{ mm}$$

Shear Stress [τ1]:

$$= Vs / (0.6 \cdot tx1 + 0.49 \cdot L43T)$$

$$= 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 0.0)$$

$$= 1637.309 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= Vs / (0.6 \cdot tx2 + 0.49 \cdot L41T)$$

$$= 3430.977 / (0.6 \cdot 349.25 + 0.49 \cdot 7.071)$$

$$= 1610.678 \text{ kgf/cm}^2$$



Shear Stress [τ_3]:

$$\begin{aligned} &= P/2(R_n + t_n)/(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot R_{L43T}) \\ &= 194.76/2 \cdot (26.5 + 6.5)/(0.49 \cdot 7.071 + 0.6 \cdot 698.5 + 0.49 \cdot 0.0) \\ &= 7.605 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(1637.3, 1610.7, 7.6) \\ &= 1637.3 \text{ must be less than or equal to } 1680.4 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -25 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: B
From Figure 3.7M- Non PWHT, the MDMT is	: -1 °C
Temperature reduction per Graph 3.12	: 24 °C
The computed stress ratio is	: 0.394

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*)/(t_{nom} - c_i - c_{ext}) \\ &= (2.562 \cdot 1.0)/(8.0 - 1.5 - 0.0) \\ &= 0.394 \end{aligned}$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -25 °C

The Drop for this Nozzle is : 0.0508 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1000.0508 mm

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